IYPT Pinhole Sunglasses

Theoretical Handbook 3

Consolidated: Single-Aperture Theory, Array Design, Ghosting, Optimization, and Vignetting

Consolidates Theoretical Handbook 2 (single-aperture) through the `array_optimization/` pipeline and Chapter 13's vignetting verification

Contents

Preface **3**

Executive Summary **4**

1 Problem Statement and Requirements **5**
1.1 The official problem . . . 5
1.2 Requirements R1–R7 . . . 5

2 Single-Aperture Theory and Optimization **6**
2.1 The architectural fix: two-stage propagation, not a plain FFT . . . 6
2.2 Governing equations . . . 6
2.3 Full formula catalog . . . 6
2.4 The critical architectural correction . . . 7
2.5 Worked numerical examples . . . 8
2.6 Processing-plan completion status . . . 8
2.7 Key single-aperture results . . . 8
2.8 R7: acuity translation status . . . 8
2.9 Patent cross-reference . . . 9
2.10 The brightness-constrained optimum, in full . . . 9
2.11 Percentile-choice robustness . . . 10
2.12 Citation discipline . . . 11
2.13 Photometric chain, MTF/CSF, and three independent acuity models . . . 12
2.14 Sensitivity, validation, and Pareto fronts . . . 12
2.15 Model comparison: why the numbers disagree, diagnosed three ways . . . 12
2.16 Two confirmed anomalies, investigated rather than assumed away . . . 17

3 Array Design: Ghosting Theory **19**
3.1 Motivation: a contradiction between the original formula and real patents . . . 19
3.2 Literature grounding . . . 19
3.3 Reconciliation against real products . . . 20
3.4 The ghost-separation formula: found wrong, corrected, and independently validated twice . . . 20
3.5 The continuous risk model, replacing binary Regime A/B/C . . . 20
3.6 Harmonic series: what actually turned out to matter . . . 21
3.7 The N_{eff} alignment resolution problem (F3) . . . 22

4 Array Optimization Pipeline **24**
4.1 Defect 1: $r_{max,lens}$ excluding 100% of the sweep . . . 24
4.2 Defect 2: ghost-formula validation, and confirming the wiring . . . 24
4.3 The feasible region, before optimization . . . 25
4.4 Packing geometry: why hex . . . 25
4.5 Weight sensitivity, in full . . . 26

4.6 Remaining gaps in the geometric model . . . 26
4.7 Smaller consistency findings (F5, F6) . . . 27
4.8 Pipeline architecture . . . 27
4.9 Why a Pareto front, not a weighted score . . . 27
4.10 The 274→90→20 funnel . . . 28

5 Vignetting: Chapter 13 Implemented and Verified **30**
5.1 The geometry . . . 30
5.2 Implementation and validation . . . 30
5.3 Field-averaging result: the 20 candidates, static vs. gaze-integrated . . . 31

6 Honest Open Items **32**
6.1 R7 acuity model: parked, single-aperture only . . . 32
6.2 Unmodeled physiological effects (single-aperture error budget) . . . 32
6.3 Transmission-target scope gap . . . 32
6.4 Single-aperture grid resolution staleness . . . 32
6.5 Manufacturing tolerance . . . 33
6.6 $w(\varphi)$, the gaze-usage distribution . . . 33
6.7 No experimental confirmation . . . 33
6.8 No single resolved answer via a weighted objective . . . 33

7 Roadmap and Next Steps **34**
7.1 Improvements identified but not all yet built, in priority order . . . 34

A Full Recommended-Candidate Table **35**

B Symbol Table **36**

C Module and Code Inventory **37**

D Reproducibility: Key Commands **38**

Preface

This handbook consolidates, rather than re-derives, this project's existing record: the single-aperture theory and results already written up in Theoretical Handbook 2, the array-design/ghosting/optimizati work in `Array_Design/` (`ANALYSIS.md`, `RECOMMENDATION.md`, `ghosting/THEORY.md`, `ghosting/RESULTS.md`), and the `array_optimization/` pipeline's own audit trail (`GAPS_CLOSED.md`, `README.md`) including the Chapter 13 vignetting verification. Every number below is read from one of those sources or a CSV it produced, not recalled from memory – where a number could not be found in an existing file, this is stated rather than approximated. This document does not replace any of its sources; each chapter names the file it condenses, for anyone who needs the full derivation.

Three conventions carry over from Handbook 2 unchanged. First, every reported number is either a first-principles calculation, a result read directly from this project's own simulation output, or an honestly-labelled citation – nothing is presented with more certainty than its basis supports. Second, where two models or two passes disagree, this handbook diagnoses the disagreement to a specific mechanism rather than picking a winner silently. Third, what is *not* yet known or built is recorded as plainly as what is (Chapter 6) – an honest account of the project's current boundary, not an admission to minimize.

Executive Summary

Single aperture (Ch. 2). Pure blur minimization places the optimum aperture at 300–400 μm across 1–8D myopia – below the verified clinical bracket (0.94–1.75mm). A physically modest brightness floor (6–9% of full natural-pupil illuminance) moves the optimum into that bracket (1000–1250 μm), uniformly across myopia and independent of which encircled-energy percentile is used. The deprecated closed-form formula chain diverges from the exact numerical result by 13.8% (cornea plane) to 64.8% mean/192.8% max (retina plane, where it is not merely imprecise but physically wrong). The literal acuity translation needs a core-width metric (FWHM), not $r_{EE}(50\%)$, which overstates real clinical acuity loss by roughly 7×.

Array design (Ch. 3). The original ghost-separation formula does not vanish for an emmetropic eye and was replaced with one that does, validated two independent ways (R^2= 0.995 vs. an independent measurement; $R^2 = -1701.7$ for the original). Under the corrected formula, the dangerous harmonic is n=1 at 20/20 dense points tested – not a higher harmonic, contrary to an earlier, superseded finding. The binary Regime A/B/C ghosting gate is replaced with a continuous risk model after finding it mislabels designs with 60–99% two-or-more-hole occupancy as "safe."

Array optimization (Ch. 4). A grid-resolution defect excluding 100% of a 1370-point blur sweep was found, fixed, and reversed a real conclusion (hex's previously-"optimal" $M \geq 4$D design accepted real ghosting risk; corrected, it matches square's safe design). The final pipeline narrows 274 candidate designs to 90 structurally safe ones to 20 on the Pareto front, none resting on a statistical (rather than structural) safety margin.

Vignetting (Ch. 5). Chapter 13's pupil-hole overlap geometry was implemented, validated against an independent Monte Carlo estimate (converging to 0.177 percentage points of full hole area), and applied to the 20 candidates: gaze-angle field-averaging over ±25° changes zero of the 20 rankings, under both an unsourced placeholder and a literature-grounded gaze distribution.

What remains open (Ch. 6): no experimental confirmation of any array-specific prediction; R7's acuity correction is single-aperture only; no numeric R2 transmission target is fixed anywhere in this project's own record; $w(\varphi)$ is a saccade-amplitude proxy, not a direct dwell-time measurement; manufacturing tolerance on hole diameter is unchecked.

Chapter 1

Problem Statement and Requirements

1.1 The official problem

> "Investigate the optimal arrangement of pinholes in an opaque film to produce sunglasses that reduce light intensity and help correct short-sightedness." (IYPT 2027)

This is a dual-objective design problem, not a single-variable one: hole diameter primarily governs optical correction, while hole count and open-area fraction govern light attenuation. Condensed from `IYPT_Pinhole_Sunglasses_Problem_Statement.md`.

1.2 Requirements R1–R7

ID	Requirement	Status, this handbook
R1	Minimize retinal blur for a myopic eye at distance	Closed: two-stage ASM propagation (Ch. 2), $r_{EE}(P)$ swept per myopia level
R2	Achieve a light-attenuation target	Whole-lens T computed per candidate (Ch. 4); no numeric target (OD/%T) is fixed anywhere in this project's own record – see §6.3
R3	Avoid visible ghosting between holes	Closed: continuous risk model replaces the binary Regime A/B/C gate (Ch. 3)
R4/R5	Field of view / gaze-angle performance	Closed for R5's geometric component: Chapter 13 vignetting, implemented and validated (Ch. 5)
R6	Experimental validation	Not built – no array-specific prediction in this project has been checked against a physical sample (§6.4)
R7	Connect blur to a visual-acuity claim	Partial, single-aperture only – parked, see §2.8

Chapter 2

Single-Aperture Theory and Optimization

Condensed from Theoretical Handbook 2's 19 chapters (`IYPT_Pinhole_Handbook_2_source.tex`).

2.1 The architectural fix: two-stage propagation, not a plain FFT

The original blueprint's error: a single FFT after the lens phase gives the field at the lens's own back focal plane, not the retina – for a myopic eye $v_r \neq v_f$, so a plain FFT silently computes a diffraction pattern with zero defocus in it, the opposite of what the problem needs. The fix: *another* Fresnel/angular-spectrum propagation, over the actual physical distance v_r, from lens to retina:

$$\text{ASM (gap)} \to \text{pupil clip} \to \text{lens phase} \to \text{ASM}(v_r) \to |\cdot|^2.$$

This makes geometric blur and diffraction blur fall out of one physically exact propagation, rather than being separately estimated and recombined by root-sum-square (RSS) – superseding the RSS combination entirely, not merely improving it.

2.2 Governing equations

The Angular Spectrum Method (ASM) computes the field $U(x, y, z)$ at propagation distance z from a known field $U_0(x, y)$ exactly, at any Fresnel number, unlike the paraxial Fresnel approximation:

$$U(x,y,z) = \mathcal{F}^{-1}\Big\{\mathcal{F}\{U_0(x,y)\} \cdot H(f_x, f_y; z)\Big\}, \qquad H(f_x, f_y; z) = \exp\Big(ikz\sqrt{1-\lambda^2(f_x^2+f_y^2)}\Big) \tag{2.1}$$

A Matsushima band-limiting mask suppresses spectral content beyond the frequency at which a plane wave, propagated the given distance, would exit the finite simulation window – a deliberate, quantified anti-aliasing measure, not a physical loss mechanism. The Fresnel number

$$N_F = \frac{a^2}{\lambda z} \tag{2.2}$$

classifies the regime: $N_F \ll 1$ Fraunhofer (far field), $N_F \sim$ 1–10 Fresnel transition (prominent near-field rings), $N_F \gg 10$ approaching geometric optics. This project's sweeps span well under 1 to over 20.

2.3 Full formula catalog

Condensed from Handbook 2's own numbered catalog (its Ch. 2), retained here because Ch. 3's ghosting-formula audit and this chapter's own divergence numbers (§2.7) both depend on comparing against the deprecated forms directly, not just the corrected ones.

#	Formula	Status
1	$1/v_f = 1/f - 1/u$ (thin lens)	Solid, unmodified throughout
2	$N_{F,\text{gap}} = R_{\text{hole}}^2/(\lambda d_{\text{gap}})$	Regime-selector validation gate
3	$a_{\text{eff}} \approx \sqrt{a_{\text{hole}}^2 + (2.44\lambda d_{\text{gap}}/a_{\text{hole}})^2}$	**DEPRECATED** – heuristic only, 15–20% error near the true optimum
4	$a_{\text{stop}} = \min(a_{\text{eff}}, D_{\text{pupil}})$	Logic solid; inherits #3's error
5	$r_{\text{geo}} = (a_{\text{stop}}/2) \cdot \lvert v_r - v_f\rvert/v_f$	Solid, paraxial; superseded as a *separate* step by the two-stage ASM architecture (§2.4)
6	$r_{\text{diff}} = 1.22\lambda v_r/a_{\text{stop}}$	Solid but incomplete – a null radius, not an energy-based size
7	$r_{\text{total}} = \sqrt{r_{\text{geo}}^2 + r_{\text{diff}}^2}$ (RSS)	**DEPRECATED, physically wrong** – diverges 64.8% mean / 192.8% max from the exact result (§2.7)
8	$a^* = \sqrt{2.44\lambda v_r v_f/\lvert v_r - v_f\rvert}$	**DEPRECATED, superseded** – inherits #7's error
–	Airy: $I(r) = I_0[2J_1(x)/x]^2$, $x = \pi a r/(\lambda v_r)$	Exact, far-field validation limit only
–	$EE(x) = 1 - J_0^2(x) - J_1^2(x)$	Exact closed form; at the first null (x=3.8317), $EE \approx 0.838$ – 16% of energy sits outside the first ring
–	$r_{EE}(P) = EE^{-1}(P)$	**Working blur metric**, replaces #6, reported at P=50, 70, 80, 90%
–	$I(0,z) = 4I_0 \sin^2(\pi N_F/2)$	Exact on-axis, oscillatory – confirms #3 is inadequate near this project's working N_F
–	$\Delta r = s v_r/d_{\text{gap}}$ (ghosting, legacy)	**DEPRECATED, wrong** – see Ch. 3
–	$T = N\pi a_{\text{hole}}^2/4 \,/\, A_{\text{total}}$ (photometry)	Solid, area-fraction definition

2.4 The critical architectural correction

Handbook 2's reference material identifies one architectural error as mattering more than any single formula: propagating lens→retina via a single FFT implicitly assumes the retina sits exactly at the lens's own focal plane – wrong whenever $v_r \neq v_f$, the entire definition of myopia. The fix is a *second* ASM propagation (Eq. 2.1), over the actual physical distance v_r. Consequence: geometric blur (Formula 5) is no longer separately computed and combined – it falls out automatically from propagating the wrong-focal-length wavefront the correct distance – and the RSS combination (Formula 7) is not merely inaccurate but structurally unnecessary, since the two blur sources were never separated to begin with.

2.5 Worked numerical examples

Fresnel number, a_{hole}=300 μm, λ=550nm, d_{gap}=12mm: $N_F = (1.5\times10^{-4})^2/[(5.5\times10^{-7})(1.2\times10^{-2})] \approx 3.41$ – squarely in the Fresnel transition regime.

Axial myopia model, P_0=60D, n'=1.336, M=4D: emmetropic axial length $L_0 = n'/P_0 = 22.267$mm; myopic $L_{\text{ax}} = n'/(P_0 - M) = 23.857$mm; $\Delta L = 1.59$mm. Eye power is unchanged between the two cases – only the lens-to-retina distance grows.

Relative illuminance, a_{hole}=1000 μm, D_{pupil}=4000 μm: $E_{\text{rel}} = (a_{\text{hole}}/D_{\text{pupil}})^2 = 6.25\%$ – not a coincidental round number, this is exactly why T_{min}=6.25% was swept as a brightness-floor test value.

2.6 Processing-plan completion status

#	Step	Status (Handbook 2)
1	Validate reduction to ideal Airy/EE at d_{gap}=0, v_r=v_f	COMPLETE, 2% (r_{EE} 50/90%), 8% (r_{EE} 80%, ill-conditioned near the null)
2	Single-hole R1 sweep	COMPLETE and extended, 75–2000 μm, 7 myopia levels
3	Cross-check vs. deprecated formulas	COMPLETE, §2.7
4	Multi-hole array (R2+R3)	Handbook 2: NOT STARTED → **this handbook, Ch. 3–4: closed**
5	Gaze-angle/vignetting (R4/R5)	Handbook 2: NOT STARTED → **this handbook, Ch. 5: closed for the geometric component**
6	Polychromatic pass	PARTIAL – representative 3-wavelength subset exists, full incoherent-sum composite does not, unchanged by this handbook
7	Acuity translation (R7)	PARTIAL, correction identified (§2.8), still single-aperture only (§6.1)
8	Experimental correlation (R6)	DEFERRED, unchanged (§6.4)

2.7 Key single-aperture results

2.8 R7: acuity translation status

Handbook 2's own requirement-status table records R7 as **PARTIAL, correction identified**: the literal translation $\theta_{\text{blur}} = r_{EE}(P)/v_r$, compared against ~ 1 arcmin, was tested against real clinical data and found to need FWHM rather than $r_{EE}(50\%)$ as the working radius – $r_{EE}(50\%)$ is roughly 7× too pessimistic against the clinical study used, while FWHM is roughly 1.6× off. This correction is single-aperture only; it has not been extended to array designs with $N_{\text{eff}} > 1$ anywhere in this project (see §6.1).

Quantity	**Value**	**Source**
Blur-only optimum aperture ($r_{EE}(50\%)$), M=1–4D	300 μm	Handbook 2 §R1
Blur-only optimum aperture ($r_{EE}(50\%)$), M=5–8D	400 μm	Handbook 2 §R1
Brightness-constrained optimum (all metrics converge)	1000–1250 μm	Handbook 2
Verified clinical bracket	0.94–1.75mm	Handbook 2
Old a_{eff} formula divergence (cornea plane)	13.8% mean, 23.7% max	Handbook 2 Finding 1
Old RSS r_{total} divergence (retina plane)	64.8% mean, 192.8% max	Handbook 2 Finding 1

Table 2.3: Headline single-aperture numbers, read directly from Handbook 2's own "Key Numbers at a Glance" table.

2.9 Patent cross-reference

KR 10-2081511's effective aperture is 0.9mm; CN 201716487U's is $<$ 0.4–0.8mm – both inside or near the diffraction-relevant bracket the sanity-check data in the problem statement describes (significant blurring below 0.75mm, negligible above 2.0mm).

2.10 The brightness-constrained optimum, in full

The 300–400 μm blur-only optimum (§2.7) sits below the verified clinical/commercial bracket (0.94–1.75mm). Rather than treating this as evidence the diffraction physics is wrong, the hypothesis tested: does the gap close once retinal brightness – entirely absent from the pure blur-minimizing objective – is added as a constraint? For a single aperture directly in front of the eye,

$$E_{\text{rel}} = \left(\frac{\min(a_{\text{hole}}, D_{\text{pupil}})}{D_{\text{pupil}}} \right)^2, \qquad a^*_{\text{practical}}(M, T_{\min}) = \underset{a_{\text{hole}}:\, E_{\text{rel}}(a_{\text{hole}}) \geq T_{\min}}{\arg\min} r_{EE}(50\%) \tag{2.3}$$

deliberately distinct from the array photometry formula (§2.3) – follows directly from energy conservation, since diffraction redistributes power spatially but does not destroy it. At $T_{\min}$=0 this exactly reproduces the unconstrained result.

brightness floor $T_{\min}$	**constrained optimum a^***	**vs. clinical bracket**
0% (unconstrained)	300–400 μm	below
1%	400 μm	below
3%	700 μm	below
6.25%	1000 μm	**inside**
9%	1250 μm	**inside**
12.25%	1500 μm	upper edge

Table 2.4: A brightness floor of roughly 6–9% of full natural-pupil illuminance – physically modest, comparable to ordinary sunglass tinting – moves the optimum into the clinical bracket uniformly across every myopia level tested. This is Handbook 2's headline single-aperture reconciliation result, cited throughout §2.7.

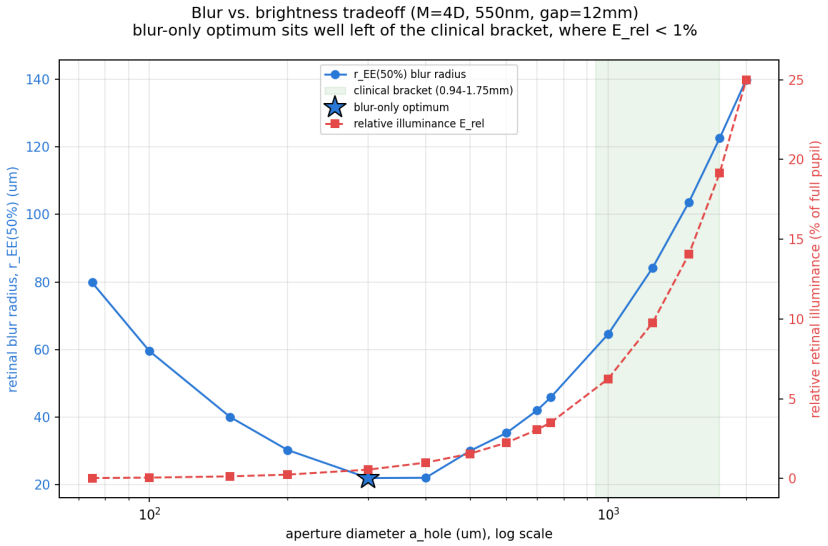


Figure 2.1: Blur radius and relative retinal illuminance vs. aperture, M=4D. `stage3a_fig1_blur_vs_brightness.png`.

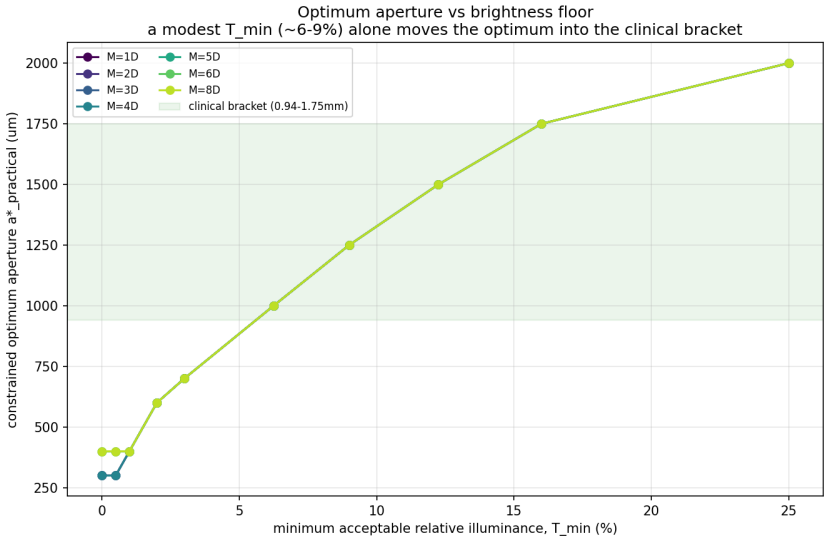


Figure 2.2: Constrained optimum aperture vs. $T_{\min}$, all myopia levels; green band is the clinical bracket. `stage3a_fig2_optimum_vs_Tmin.png`.

2.11 Percentile-choice robustness

$r_{EE}(70\%)$, extracted directly from the raw retinal field with no pipeline modification, was combined with the pre-existing 50/80/90% columns. Unconstrained, the four percentiles disagree modestly (300–400 μm for 50/70%; 400–600 μm for 80/90%). Once the 6.25% brightness floor is applied, **all four percentiles converge to exactly the same answer – 1000 μm – at every myopia level**: the brightness floor matters far more than which encircled-energy percentile is chosen.

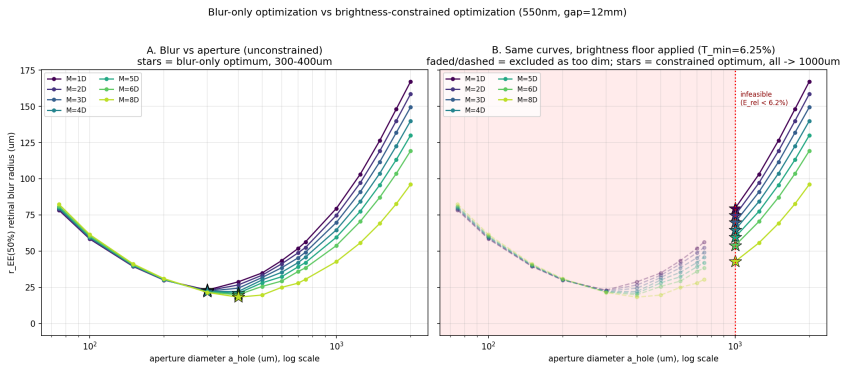


Figure 2.3: Panel A: unconstrained blur-vs-aperture. Panel B: the same curves with the 6.25% floor applied. `stage3a_fig4_blur_then_constraint.png`.

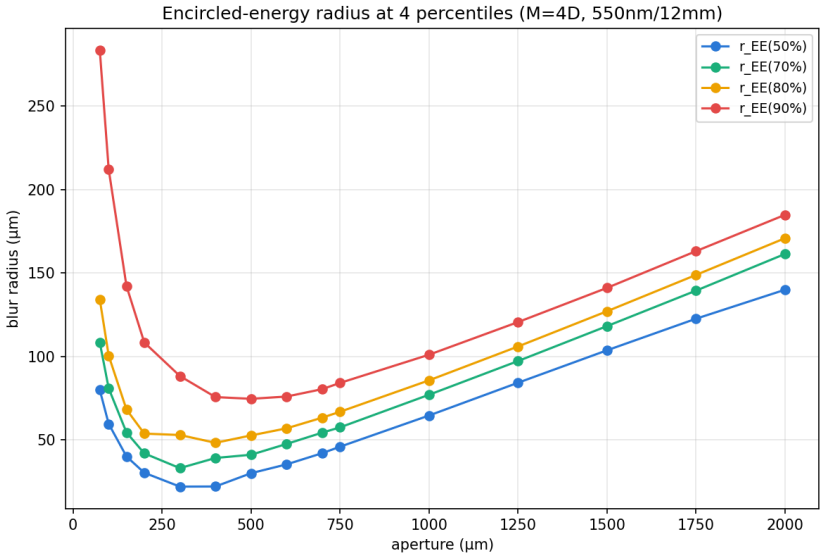


Figure 2.4: r_{EE} at 50/70/80/90% vs. aperture, M=4D. `ee_all_percentiles.png`.

2.12 Citation discipline

Every implemented relationship in Handbook 2's comprehensive metric suite is classified as *first_principles* (needs no citation – energy conservation, the Fourier-optics definition of MTF), *published* (a specific named model, with any simplification stated), or *heuristic* (a simplified stand-in, e.g. where the full multi-parameter Barten CSF could not be reproduced exactly). Twenty citation entries cover: the Strehl ratio, MTF/OTF, the Airy diameter, F-number, RMS spot radius, the troland unit and CIE photopic luminous efficacy, two CSF parametrizations, the Rose–DeVries law, Weber contrast, the Blackwell (1946) foveal-detection dataset, MAR/Snellen conversion, the Visual Strehl metric, the Stiles–Crawford effect, and ocular aberration magnitudes. This handbook's own new material (Ch. 3–5) follows the same discipline: the Barten (1999) CSF (§3.5) and the saccade-amplitude/center-bias literature (§5.3) are both *published*, explicitly distinguished from the *heuristic* `uniform` gaze-distribution placeholder they sit alongside.

2.13 Photometric chain, MTF/CSF, and three independent acuity models

The full chain from scene luminance to retinal photon flux reuses the standard image-irradiance camera equation, $E_{\text{retina}} = \pi N \tau L_{\text{scene}}/[4(\text{F}\#)^2]$, with heuristic ocular transmission τ=0.90 (range 0.85–0.95) exposed as an overridable argument. MTF is computed exactly (no free parameters) as $|\mathcal{F}\{\text{PSF}\}|$, radially averaged. Two published CSF parametrizations are implemented and deliberately never merged – a band-pass model peaking near 4 cyc/deg, and the Mannos–Sakrison parametrization peaking near 8 cyc/deg – and their genuine peak-location disagreement is reported as a finding, not resolved by picking one (§**??** above).

Three independent acuity models are computed and compared, never averaged into one number: (1) MAR from the MTF50/MTF10 cutoff via MAR(arcmin) = $30/C$; (2) an approximate Visual Strehl (VSX-style) metric, CSF-weighting actual vs. diffraction-limited MTF; (3) a Rose–DeVries photon-noise-limited bound, MAR $\propto \Phi^{-1/2}$, calibrated to a *relative* noise-floor change, not an absolute clinical prediction.

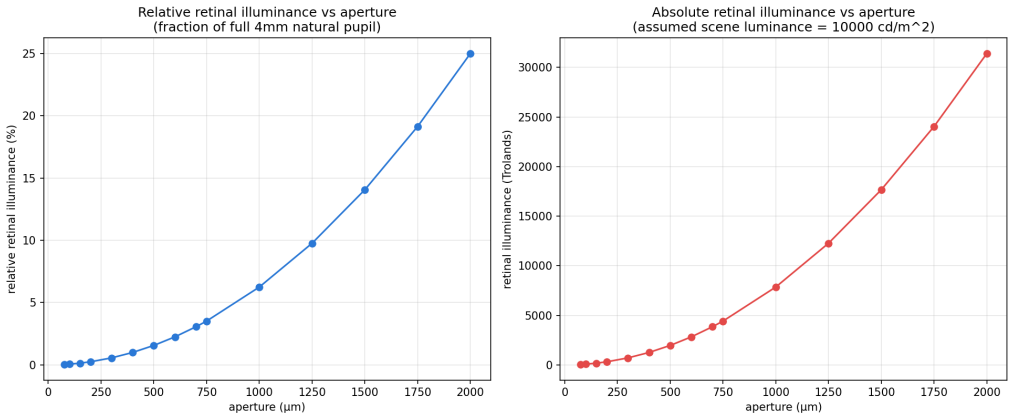


Figure 2.5: Relative retinal illuminance and absolute Trolands vs. aperture, M=4D. `illuminance_troland.png`.

2.14 Sensitivity, validation, and Pareto fronts

2.15 Model comparison: why the numbers disagree, diagnosed three ways

Handbook 2's single most important synthesis result: an independent audit comparing three implementations against each other and against real-world reference data, concluding that disagreements across this project's own history, the deprecated closed-form formulas, real clinical literature, and real patents have three distinct, individually-diagnosed causes – not one broken model.

Model	What it computes	Basis
1	Exact ASM, pinhole to cornea only. No eye, no myopia.	`stage1_v2`
2	Exact ASM×2 + lens phase: full chain with axial myopia.	`stage2`
3	The deprecated closed-form chain: a_{eff}, $r_{\text{geo}}/r_{\text{diff}}/r_{\text{total}}$ (RSS), a^*.	`classical_formulas.py`

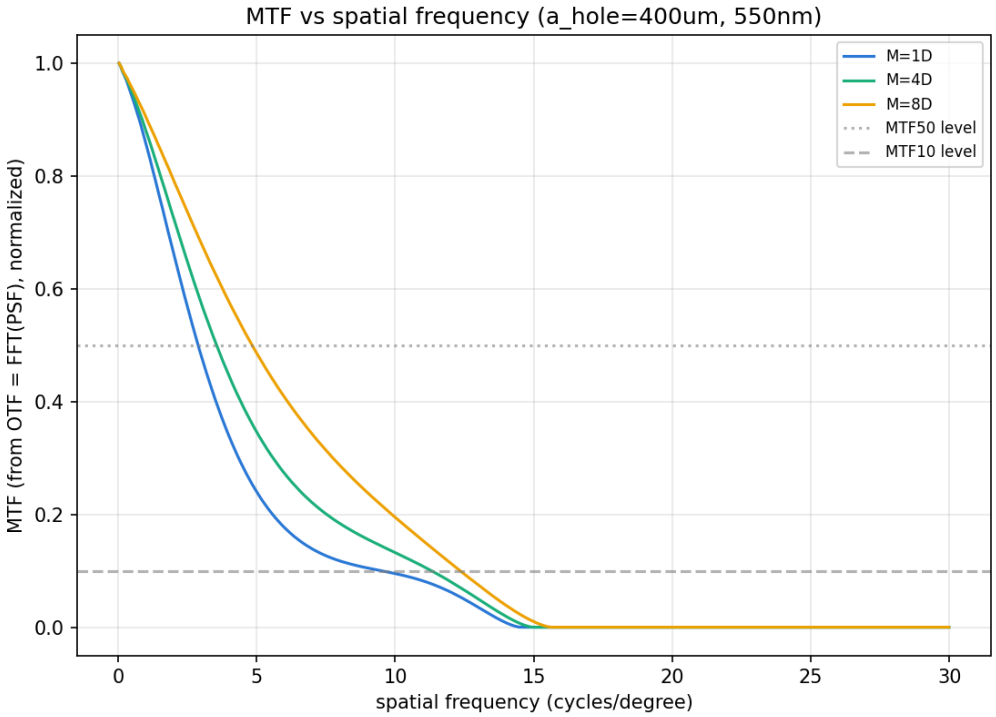


Figure 2.6: Radially-averaged MTF vs. spatial frequency, three myopia levels, MTF50/MTF10 marked. `mtf_curves.png`.

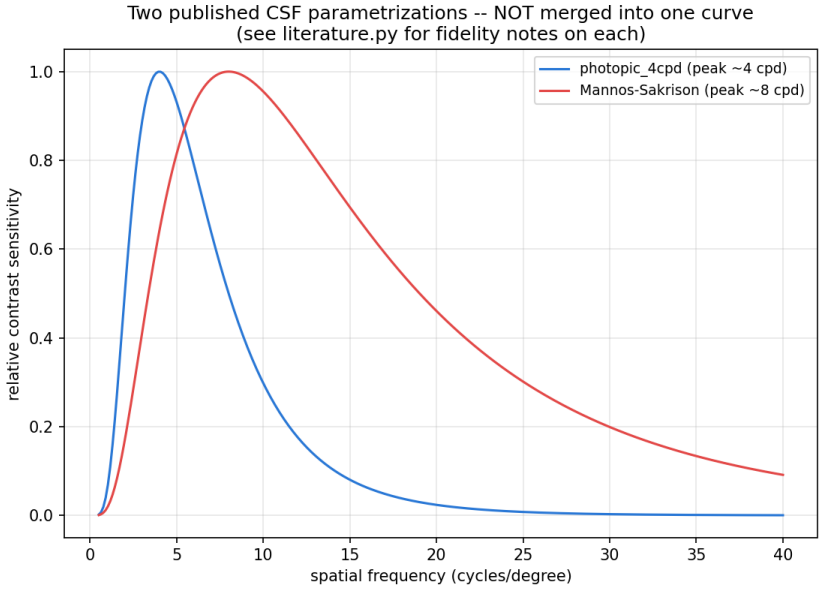


Figure 2.7: Two published CSF parametrizations, showing the genuine peak-location disagreement. `csf_comparison.png`.

Finding 1 – the old formula's accuracy depends entirely on which stage it is applied at. a_{eff} was designed as a cornea-plane spot-size estimate, and there it tracks the exact result reasonably well (Fig. 2.13) – consistently 14–24% high, same overall shape (see §2.7's table for the exact 13.8%/23.7% figures). The much larger failure occurs when the same formula family is repurposed for retina-blur optimization via RSS (§2.7's 64.8%/192.8%) – exactly the step §2.4 identifies as physically wrong. Root cause, confirmed by direct calculation:

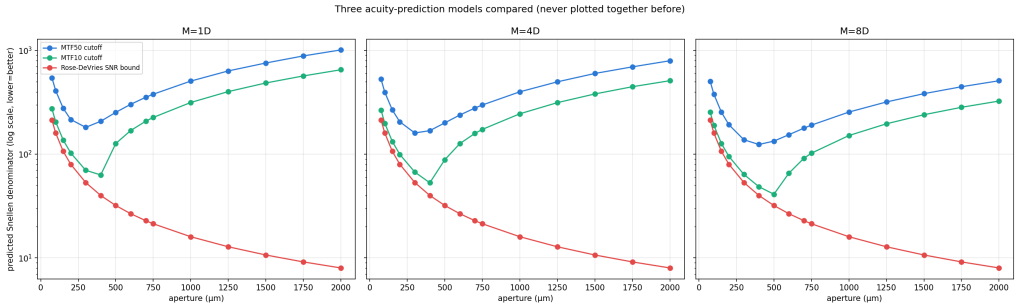


Figure 2.8: All three acuity-prediction models, per myopia level. `acuity_models_combined.png`.

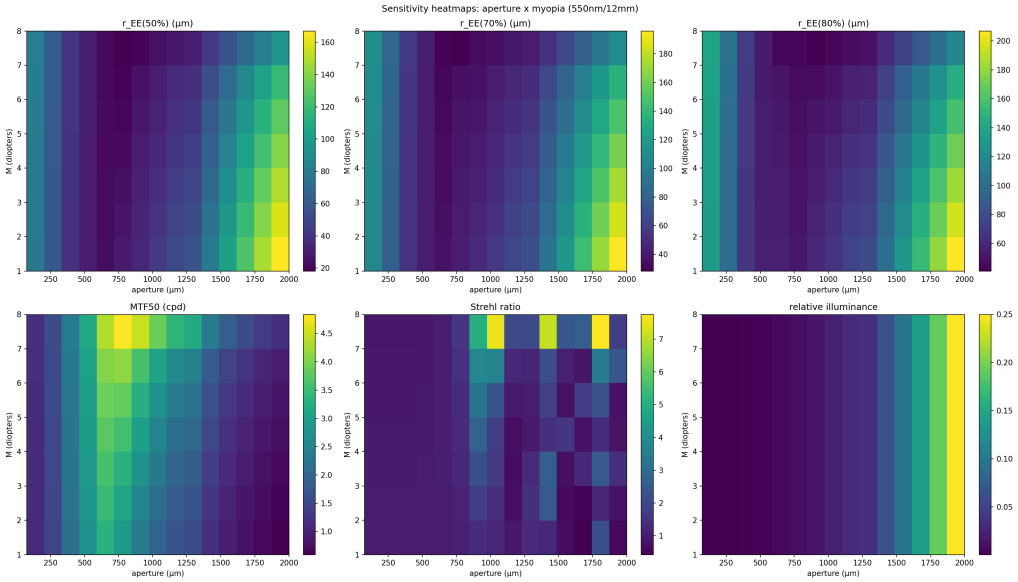


Figure 2.9: Sensitivity heatmaps: aperture vs. myopia, for $r_{EE}(50\%)$, MTF50, Strehl, and relative illuminance. `sensitivity_heatmaps_aperture_M.png`.

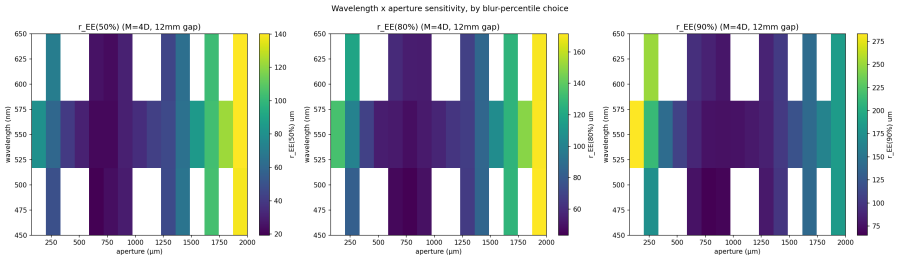


Figure 2.10: $r_{EE}(50\%)$ sensitivity: wavelength vs. aperture, M=4D. `sensitivity_heatmap_wavelength_aperture.png`.

a_{eff} is itself non-monotonic at small apertures, producing a spurious secondary dip around 100–150 μm unrelated to the true optimum (Fig. 2.14).

Finding 2 – none of the three physics models land near commercial/clinical designs. All three physics-based results cluster in the 200–700 μm range, consistently smaller than any commercial or clinical product. Diagnosed as scope, not error: every physics model here optimizes blur *alone*. A 300 μm hole passes roughly 16× less light than a 1.2mm hole (power $\propto d^2$), and none of the three compared models builds in throughput. The brightness-constrained

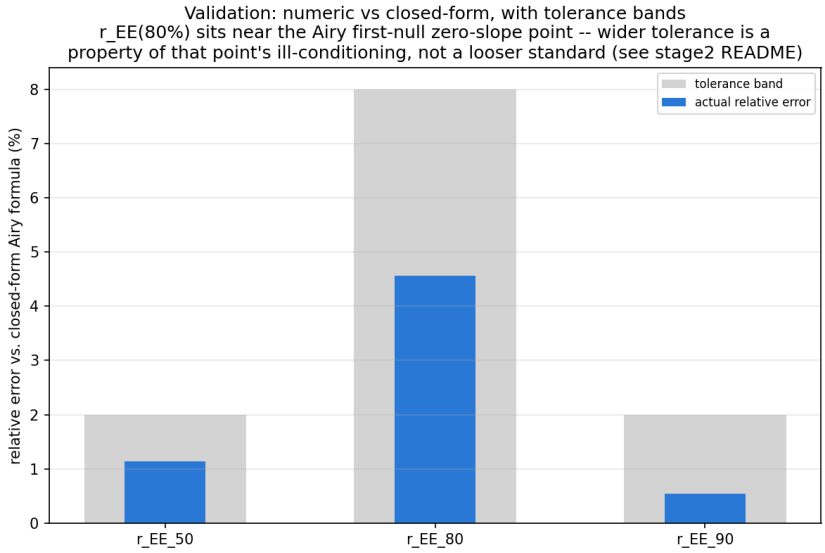


Figure 2.11: Numeric vs. closed-form Airy result, explicit tolerance bands per percentile. `validation_tolerance_bands.png`.

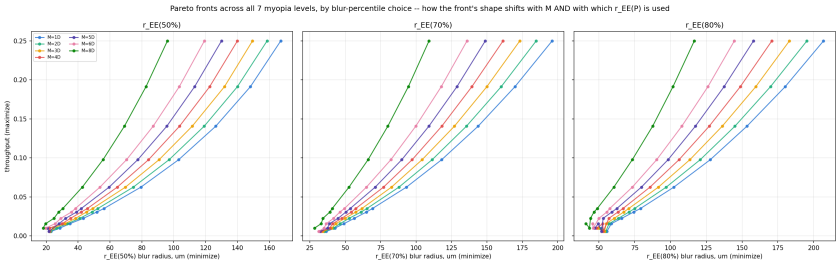


Figure 2.12: Pareto-optimal fronts (blur radius vs. throughput), overlaid across all seven tested myopia levels. `pareto_all_M_overlay.png`.

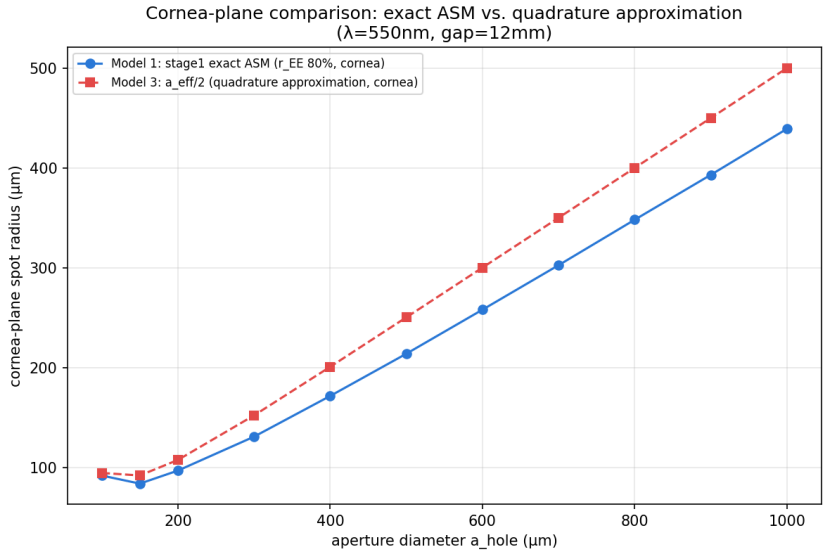


Figure 2.13: Cornea-plane: old $a_{\text{eff}}/2$ vs. exact $r_{EE}(80\%)$. Biased high but correct in shape.

optimum (§2.7, 1000–1250 μm) is exactly the corrective step this finding calls for.

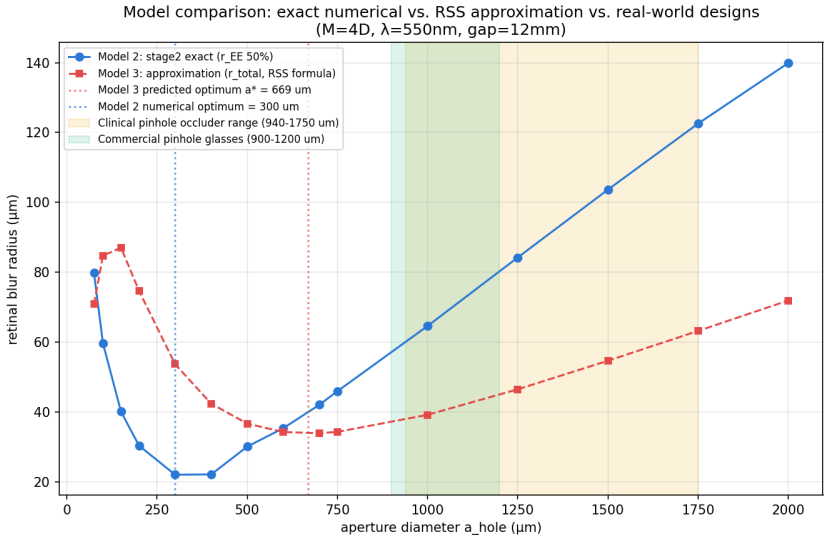


Figure 2.14: Retina-plane, M=4D: old RSS r_{total} vs. exact $r_{EE}(50\%)$, showing the RSS formula's own spurious secondary dip near 100–150 μm.

Source	**Aperture diameter**
Model 2 (this project, r_{EE} 50%)	300–400 μm
Model 3 (closed-form a^*)	473–1339 μm
Rayleigh pinhole-camera formula, $d = 1.9\sqrt{f\lambda}$	210–225 μm
Clinical pinhole occluders (optometry)	940–1750 μm ($\sim$ 1200 μm typical)
Commercial pinhole glasses	900–1200 μm

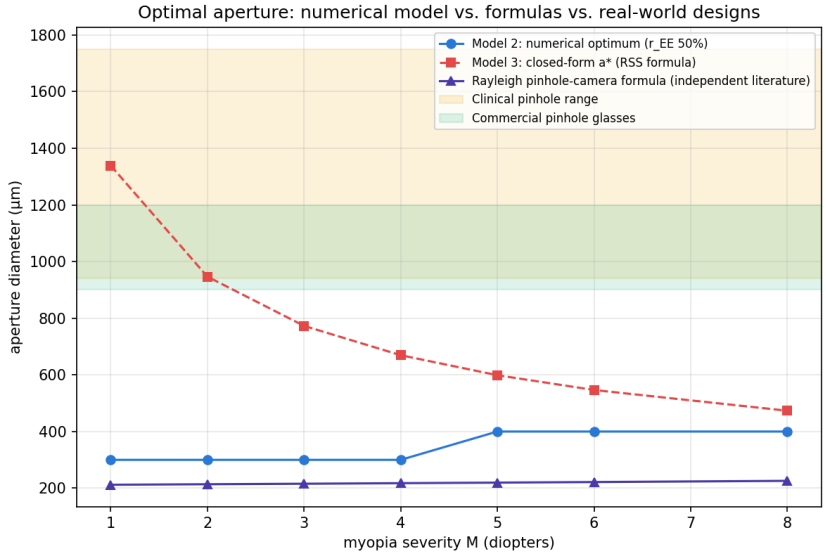


Figure 2.15: All five reference points on one axis. `comparison_optimum_vs_rayleigh.png`.

Finding 3 – a genuine metric-selection error, confirmed against a real clinical study. Applying $\theta_{\text{blur}} = r_{EE}(P)/v_r$ literally at the same condition a real clinical study measured (1.2mm aperture, $\sim$ 5D myopia, reported $\sim$ 20/30 Snellen, 0.17 logMAR): $r_{EE}(50\%)$ predicts acuity nearly **seven times worse** than the clinical measurement; FWHM lands within roughly 1.6×. Mechanism: $r_{EE}(50\%)$ is a cumulative-*energy* radius, counting diffuse peripheral light in faint outer Fresnel rings that does not meaningfully impair letter recognition, while

Metric	Value	Implied blur	Implied Snellen
$r_{EE}(50\%)$	73.8 μm	10.45 arcmin	$\sim$ 20/209
FWHM (core-width)	16.9 μm	2.39 arcmin	$\sim$ 20/48

FWHM measures only the bright, legibility-relevant core. This does not mean $r_{EE}(P)$ is wrong everywhere – for the blur-*distribution* objective it is exactly the specified metric; it is specifically wrong when repurposed, unmodified, for an acuity claim (§2.8, §6.1).

2.16 Two confirmed anomalies, investigated rather than assumed away

Retina-plane FWHM ring-hopping: FWHM jumps from 4.7 μm to 154.7 μm between two adjacent 250 μm-spaced test apertures at M=4D – over 30×. Retina-plane FWHM is not a reliable single-value metric for this system. **Strehl non-monotonicity**: a tenfold finer sweep, run specifically to test whether the coarse sweep's non-monotonic Strehl-maximizing aperture was a discretization artifact, revealed *more* local maxima (6–7, not fewer) and *larger* swings (up to 8.4×, not smaller) – the signature of genuine structure, not sampling noise.

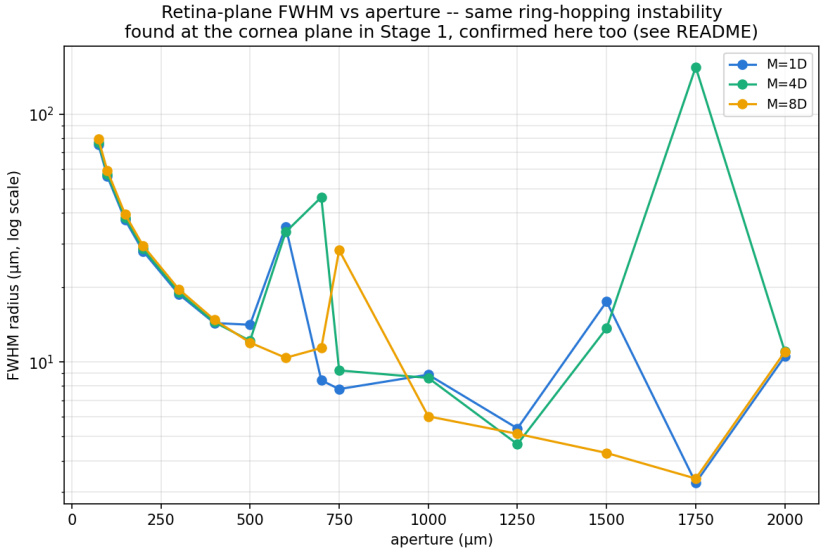


Figure 2.16: Retina-plane FWHM vs. aperture (log scale). `retina_fwhm_ring_hopping.png`.

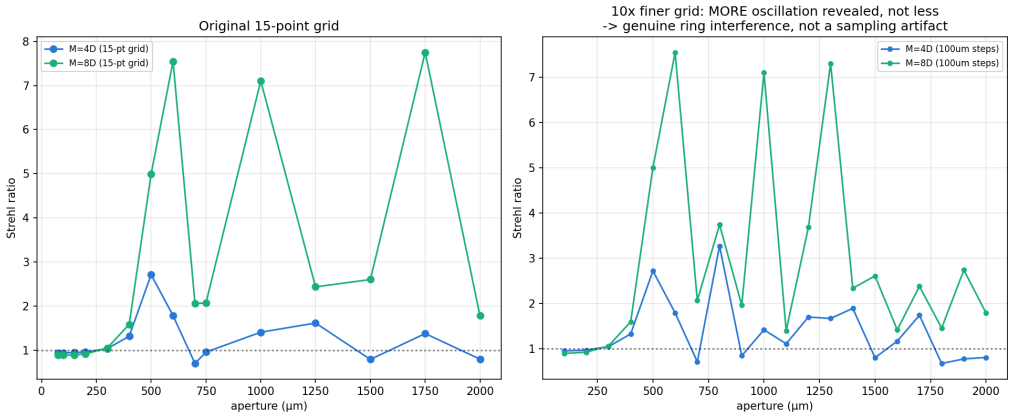


Figure 2.17: Strehl ratio at coarse vs. tenfold-finer resolution – the finer grid reveals *more* oscillation, not less. `strehl_investigation.png`.

Chapter 3

Array Design: Ghosting Theory

Condensed from `Array_Design/ghosting/THEORY.md`, `RESULTS.md`, and `ANALYSIS.md`'s F1/F2 findings.

3.1 Motivation: a contradiction between the original formula and real patents

The original criterion (`stage_array/array_photometry.py`, moved unmodified to `Array_Design/photometry/`) called a design "ghost-free" if Δr (Eq. **??**, legacy) was smaller than the single-hole blur radius $r_{EE}(P)$. Checked against two real, sold patents (KR 10-2081511, pitch 3.2–4.8mm; CN 201716487U, pitch 0.7–1.2mm; both $d_{\text{gap}} \sim 12$mm, $v_r \sim 22$–24mm), this predicted severe violations (30×–400×) for *both* – the contradiction that motivated re-deriving the formula at all.

3.2 Literature grounding

- **Gabor superlens theory** (1941/2014): the closest existing precedent for a quantitative multi-channel merging condition – mismatched pitches in a two-array relay cause channels to "unite to form superfocal lines much smaller in number than the number of microlenses." Cited as evidence this class of problem is solvable, not as a directly transferable formula (it describes a two-array relay forming a real image, not a single occluder imaged by the eye's own optics).
- **Direct clinical evidence ghosting is real**: Kim, Park, Park & Chun (2017, *J Korean Med Sci* 32(5):850–857) directly reports, in real subjects wearing real multi-pinhole glasses, "discomfort from after-images such as honeycomb images, horizontal grating, and double or triple vision," significantly worse than single-pinhole glasses – overturning the original framing (real products *do* report ghosting; the question is under what conditions it becomes severe).
- **An independent patent statement of the mechanism**: a separate US patent states, independent of anything in this project, "If two pinholes are separated by less than the diameter of the pupil aperture, two pencils of ray…form two retinal images. Further, too large of a separation will result in dead spots…while too small…will produce multiple images." A discrete precondition (are ≥ 2 holes simultaneously illuminating the pupil) – exactly the N_{eff} quantity this project's own lattice enumeration computes.
- **Periodic-artifact visibility** (display-engineering "screen-door effect"): confirms the right framework – express periodicity as cycles/degree, determine modulation depth, compare against the human CSF threshold at that frequency.

3.3 Reconciliation against real products

Design	Pitch	N_{eff}	Regime	Matches evidence?
KR 10-2081511	3.2–4.8mm	0/1.23	A	Consistent – patent never discusses ghosting
CN 201716487U	0.7–1.2mm	23–28	B (untested)	Plausible, not independently confirmed
Kim et al. 2017 MPH	pitch unstated	likely C	Danger zone	Directly matches real reported symptoms

Table 3.1: This does not cleanly vindicate both patents – it replaces one crude formula with three separately-justified mechanisms and correctly locates where real problems are reported.

3.4 The ghost-separation formula: found wrong, corrected, and independently validated twice

The formula originally in circulation,

$$\Delta r = s \cdot v_r / d_{\text{gap}} \qquad \text{(legacy, Eq. 1)}$$

does not vanish for an emmetropic eye: at $M{=}0$ it predicts ghost copies $1.86s$ apart in an eye with zero defocus, which is not physically possible – a perfectly focused eye images a distant point to one point regardless of how many holes admit light. The corrected expression,

$$\Delta r = s \cdot \frac{|v_r - v_f|}{v_f} = \frac{s \cdot M}{P_0 - M} \qquad \text{(corrected, Eq. 1}'\text{)},$$

was derived three independent ways (paraxial ray transfer, wave-optics propagation, the $M{=}0$ limit) in the original consolidation, and re-validated a second, independent time in this handbook's own pipeline (Ch. 4, Stage 0): a genuine two-hole coherent propagation plus an incoherent-sum reference, with Δr measured by peak detection, not formula lookup.

Formula	Form	R^2 vs. 32 measured points
(a) legacy	$\Delta r = s v_r / d_{\text{gap}}$	−1701.71
(b) corrected	$\Delta r = sM/(P_0 - M)$	0.9951
(c) free fit	$\Delta r = A[sM/(P_0 - M)] + B$	0.9968, A=1.041, B= − 2.75

Table 3.2: Ghost-formula validation, `array_optimization/stage0_ghost_validation/ghost_formula_valida` The free fit independently rediscovers (b) within its own pre-registered 5% consistency band.

3.5 The continuous risk model, replacing binary Regime A/B/C

`ghosting_regime.py`'s original Regime A/B/C classifier reads its worst case off the *minimum* N_{eff} over pupil alignment – correct for brightness (fewest holes = darkest) but wrong for ghosting (most holes = worst ghosting). At D_{pupil}=4mm, square packing, pitch 2500 μm is classified Regime A ("no second hole") while carrying $P(N_{\text{eff}}{\geq}2){=}0.82$ – two or more holes active 82% of the time. Only pitches above $s_{\text{solo}} \approx$ 3.94mm (square) / 4.00mm (hex) – where even the *maximum* N_{eff} is 1 – are ghost-free by geometric construction.

`array_optimization/stage2_classification/classify.py` replaces the binary gate with

$$\text{risk} = P(N_{\text{eff}} \geq 2) \times \text{worst-harmonic perceptual margin},$$

risk = 1.0 being the point where the probability-weighted margin crosses the Barten (1999) CSF detection threshold. The old binary label is still computed and exposed for reference, never gated on.

3.6 Harmonic series: what actually turned out to matter

Ghosting theory's four-pass history (`ghosting/RESULTS.md`) found, then partly un-found, a harmonic-series effect:

- Pass 1 (heuristic CSF): zero Regime C found, an artifact of a CSF model evaluated far outside its fitted range.
- Pass 2 (real Barten 1999 CSF): reverses this completely – 20/20 previously-"safe" dense points reclassified Regime C.
- Pass 3 (harmonic extension, legacy Δr): found the worst-case harmonic is *never* n=1 – up to n=20, margin 4.4× the fundamental-only estimate.
- Pass 4 (harmonic extension, *corrected* Δr): Pass 3's headline finding does **not survive**. With the corrected formula the fundamental already sits at 2.9–8.3 cpd, at or past the CSF peak, so the worst harmonic is n=1 at **20/20** dense points tested, worst-margin÷fundamental-margin = 1.000 exactly.

This project's own record therefore directly contradicts a claim that "the dangerous harmonic is usually n=2–3": under the validated formula it is n=1 essentially everywhere in this project's geometry, not a higher harmonic.

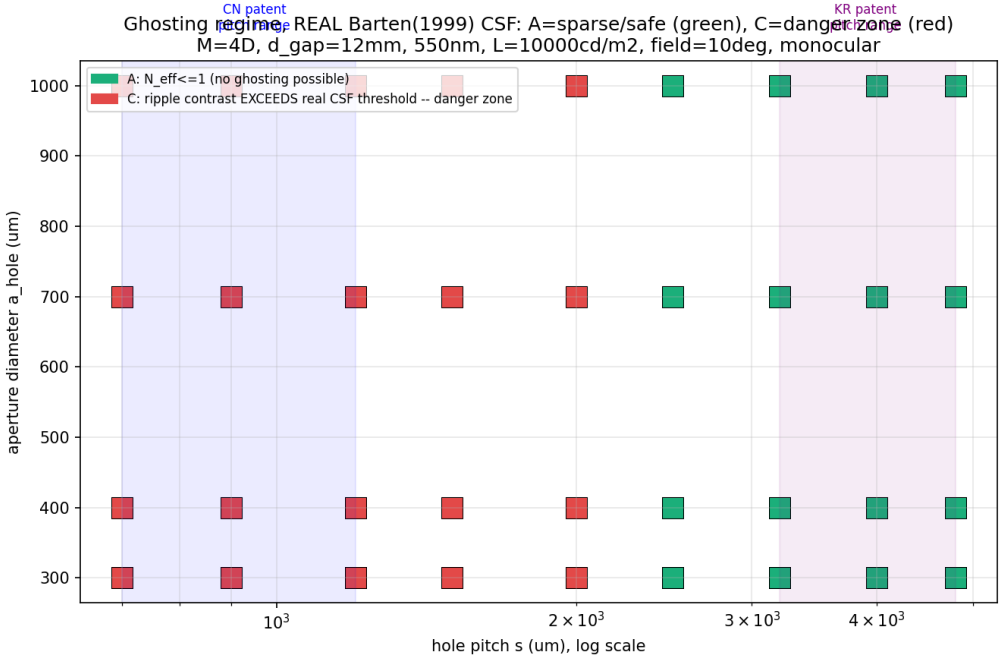


Figure 3.1: Regime map under the real Barten (1999) CSF (Pass 2). `ghosting/ghosting_regime_map_barten.png`.

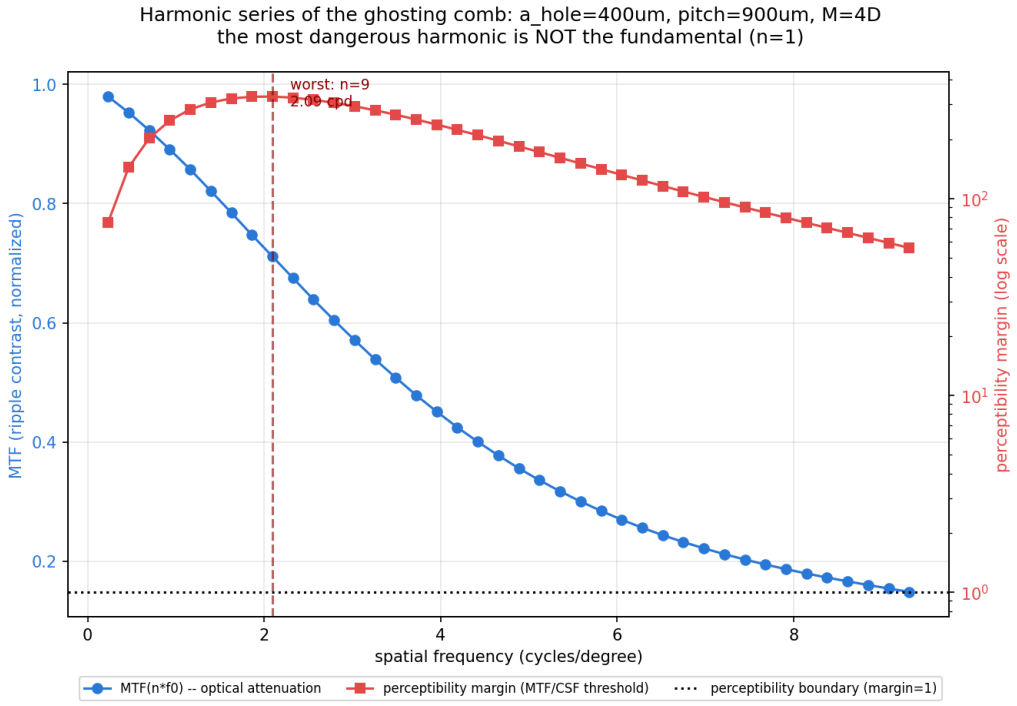


Figure 3.2: Harmonic margin profile (Pass 3, legacy formula – superseded by Pass 4's n=1 finding, kept for the historical record). `ghosting/harmonic_margin_profile.png`.

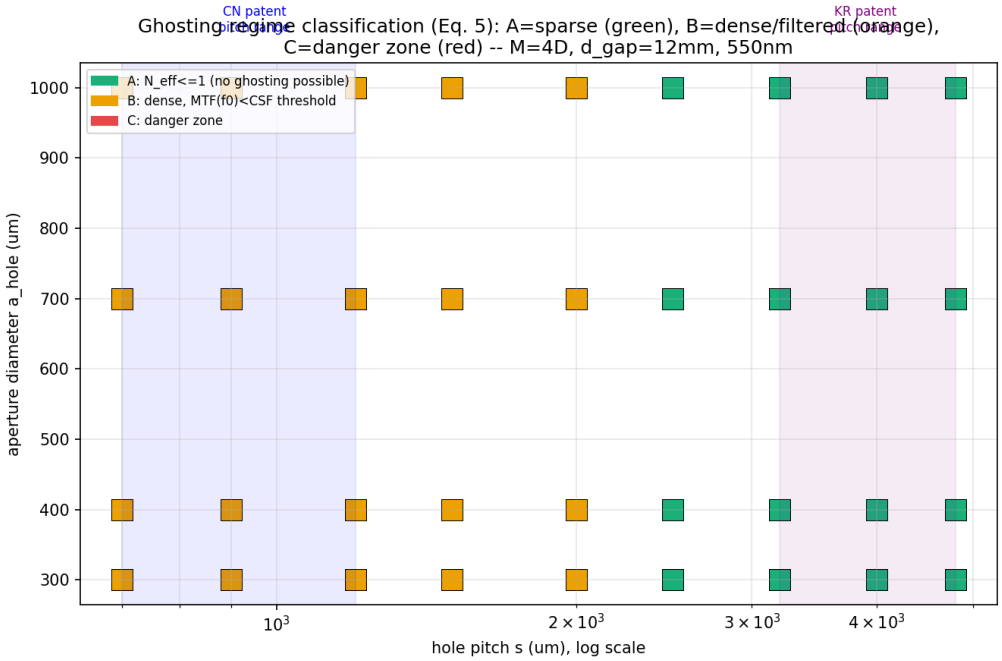


Figure 3.3: Regime map, heuristic CSF (Pass 1) – superseded by the Barten map above, kept for the historical record of why Pass 1 found zero Regime C. `ghosting/ghosting_regime_map.png`.

3.7 The N_{eff} alignment resolution problem (F3)

A second, independent problem was found alongside F1/F2: `n_holes_exact_sweep_alignment()` defaults to 25 offsets per axis; `ghosting_regime.py` calls it with 15. Both are too coarse exactly where the design boundary sits, and both err optimistic:

pitch (μm)	**quantity**	$n_{\mathbf{offsets}}$ **15/25**	$n_{\mathbf{offsets}}$ **41/61/121**
2900	min N_{eff}	**1** ("never dark")	**0** (goes dark)
3900	max N_{eff}	**1** ("never ghosts")	**2** (can ghost)

Table 3.3: The sampled offsets never land on the worst alignment (a lattice-cell corner) unless the grid is fine enough. `joint_design_space.py` uses 61 throughout (61 and 121 agree at every pitch tested); the stored CSVs in `photometry/` and `ghosting/` were produced with the coarse defaults.

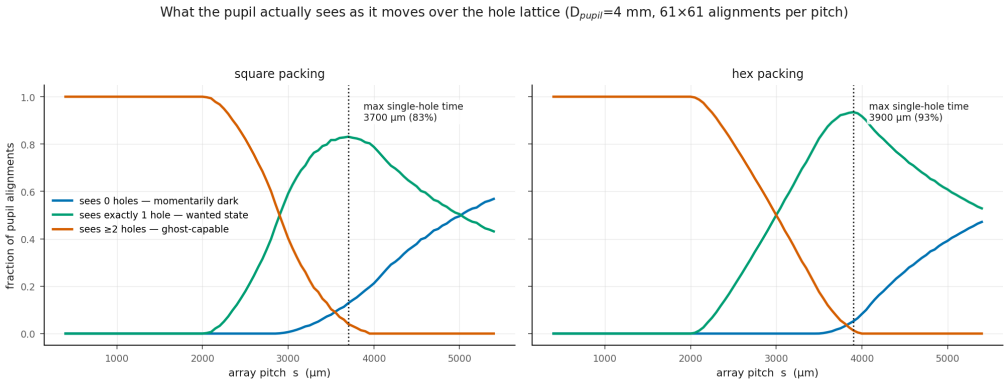


Figure 3.4: $P(0 \text{ holes})$, $P(\text{exactly } 1)$, $P(\geq 2)$ vs. pitch, the quantity F2 is built on. `Array_Design/alignment_fractions.png`.

Chapter 4

Array Optimization Pipeline

Condensed from `array_optimization/GAPS_CLOSED.md`, `README.md`, and `ANALYSIS.md`'s third-pass Addenda.

4.1 Defect 1: $r_{\max,\mathbf{lens}}$ excluding 100% of the sweep

The original multi-hole blur sweep (`multi_hole_blur_sweep.csv`, 1370 points) excluded off-axis holes at **every single point tested (100%)**, silently understating blur everywhere $N_{\text{eff}} > 1$. Diagnosed by direct instrumentation, not assumed: $r_{\max,\text{lens}}$ is a pure function of the fixed simulation grid (wavelength, focal length, pixel size) – identical across every aperture and myopia level – and was permanently smaller than the eye's 2000 μm pupil radius at the old grid resolution (grid_N=2048). Not a crop-window bug, not a units bug (both directly ruled out). Fixed by raising grid_N to 4096; re-verified with **zero exclusions** across the full 1370-point regenerated sweep.

A second, unrelated defect surfaced while regenerating at the new resolution: an unbounded PSF-field cache grew to ~134MB/entry, driving the host machine's swap and disk to capacity mid-run. Fixed with a size-capped (FIFO, 20-entry) cache; the re-run held stable memory throughout (peak under 4GB).

What this changed: the hex $M \geq 4$D reversal

Fixing Defect 1 reversed a real, previously-reported conclusion. Before the fix, hex's "optimal" design at M=4–8D accepted real ghosting risk (98–137, well past the safe threshold of 1.0); after, it matches square's safe, zero-risk design:

M (D)	packing	a (μm)	s (μm)	N_{eff}(worst)	risk (before→after)	blur×single
4	hex	2000	2200→**4200**	1→**0**	136.6 → **0.0**	4.1×
6	hex	2000	2200→**4200**	1→**0**	116.4 → **0.0**	5.4×
8	hex	2000	2200→**4200**	1→**0**	98.3 → **0.0**	6.1×

Table 4.1: Traced to source, not just observed: `blur_ratio` at hex's old pick roughly *doubled* (M=4D: 4.12 → 8.25×) once previously-excluded off-axis holes were correctly included – `risk` itself is numerically identical before and after; the ghosting classification was never wrong, the blur feeding the joint cost function was.

4.2 Defect 2: ghost-formula validation, and confirming the wiring

Covered in §3.4. `ghosting_risk.py`'s runtime wiring was unit-tested, not assumed from a parameter's name, and found to already call the validated corrected formula's own function exactly – no code change was needed there, only verification.

4.3 The feasible region, before optimization

Before any scored optimum existed, `joint_design_space.py` produced a *feasible region* – deliberately not collapsed to a single (diameter, pitch) pair, since no joint optimisation had yet been run (G1). Conditioned throughout on D_{pupil}=4mm, d_{gap}=12mm, λ=550nm, P_0=60D, axial myopia, on-axis viewing: **Which constraint bounds which edge**: diameter's lower edge is

	M=1**D**	M=4**D**	M=8**D**
hole diameter	700–1500 μm	400–600 μm	300–400 μm
pitch (square)	3.94–4.4mm (myopia-independent)		
pitch (hex, preferred)	4.00–4.5mm (myopia-independent)		
blur r_{EE}(50%), one hole active	$\sim$ 7.3 μm	$\sim$ 12.0 μm	$\sim$ 17.9 μm
retinal illuminance, hex mean	2.3–12.9%	0.7–2.1%	0.4–0.9%
P(0 holes)	hex 8–26%, square 18–35%		

Table 4.2: The diameter axis moves with myopia; the pitch axis does not, because pitch bounds come from pupil geometry (how many holes fit in a 4mm circle), which does not know about refractive error – the single most useful structural result of putting all three constraints on one plot.

diffraction (smaller holes blur more); its upper edge is geometric defocus (larger holes admit a wider defocused pencil, cut at +10% tolerance above the minimum). Pitch's lower edge is *ghosting* – hard, geometric, at s_{solo}. Pitch's upper edge is *brightness* – soft, a tolerance choice, the only edge a designer has a real dial to turn.

A physical subtlety the naive transmission formula misses: $T = N\pi a_{\text{hole}}^2/4\,/\,A_{\text{total}}$ (Eq. in §2.3) treats the perforated sheet as a spatially-averaged neutral-density filter – correct only when pitch $s \ll D_{\text{pupil}}$, so many holes sit under the pupil at any instant. Checked against the two real patents examined here (KR: pitch 3.2–4.8mm; CN: 0.7–1.2mm) against a natural 3–4mm pupil: these pitches are *not* much smaller than the pupil, they are comparable to it. The actual instantaneous hole count under the pupil is therefore a small integer (often 1–4), not a smooth average, and is sensitive to the exact alignment between pupil center and hole lattice – which is exactly why N_{eff} is computed by exact discrete lattice enumeration (§3.5) throughout this handbook, rather than by the continuum area-ratio formula alone.

4.4 Packing geometry: why hex

Hex packing dominates square on every geometric measure tested (`alignment_fractions.csv`, `pupil_sensitivity.csv`), at D_{pupil}=4mm:

quantity	**square**	**hex**
s_{dark} (pitch above which the view can go black)	2877 μm	**3487 μm** (+21%)
s_{solo} (pitch above which 2 holes never co-occur)	3936 μm	4001 μm
width of the unavoidable-compromise gap	1059 μm	**514 μm** (half)
best achievable P(exactly one hole)	0.83 @ 3700 μm	**0.93** @ 3900 μm
P(0 holes) at that best pitch	0.13	**0.05**

Table 4.3: Hex wins for the geometric reason that its Voronoi cell has a smaller circumradius ($s/\sqrt{3}$ vs. $s/\sqrt{2}$), covering the pupil at a larger pitch. Caveat: no ghosting-regime sweep, MTF/CSF margin, or multi-hole PSF had ever been computed for hex when this was first found – the advantage above was packing geometry alone, subsequently closed by this handbook's Stage 0–3 pipeline running both packings through the same corrected risk model.

4.5 Weight sensitivity, in full

The joint cost function $w_{\text{blur}} \cdot \text{blur_ratio} + w_{\text{dark}} \cdot P_{\text{dark}} + w_{\text{ghost}} \cdot \min(\text{risk}, 50) - w_{\text{bright}} \cdot (E_{\text{rel}}/1\%)$ was stress-tested by varying each weight 0.25×–16× its default (w_{blur}=1, w_{dark}=5, w_{ghost}=1, w_{bright}=1) at M=4D: This instability was directly re-checked against the final 20 Pareto can-

weight varied	**argmin behaviour,** 0.25×–16×
w_{blur}	moves: $a \in \{600, 2000\}\,\mu$m, s=4200 μm fixed
w_{dark}	moves: a=2000 μm fixed, $s \in \{2200, 4200\}\,\mu$m
w_{ghost}	moves: a=2000 μm fixed, $s \in \{2200, 4200\}\,\mu$m – flips to the **dense, risky** design at 0.5× or below
w_{bright}	moves: a=2000 μm fixed, $s \in \{2200, 4200\}\,\mu$m – flips to the dense design at 2× or above

Table 4.4: `optimise_sensitivity.csv`. The M=4D/square answer flips to the dense, $N_{\text{eff,worst}}$=1, real-ghosting-risk design at 0.5× the default ghosting weight or 2× the default brightness weight – well inside any reasonable range of disagreement about how those two terms should trade off, which is exactly why Stage 3 (§4.9) does not use this weighted objective at all.

didates specifically (not merely inferred from Pareto-optimality), reusing this same cost function: at M=4, 6, and 8D, under default weights *and* every tested perturbation (w_{ghost}=0.5×, w_{bright}=2×, and both combined), the specific fragile design (a=2000, s=2200) that a weighted-objective decision rule would favour is **never** one of the 20, and never becomes one under any tested perturbation, because Stage 3's admission gates (risk < 1.0, margin ≥ ceiling) are hard thresholds on measured quantities, not on any weighted combination – a structural, not merely empirical, reason the instability cannot leak into the recommended set (`array_optimization/stage3_decision`

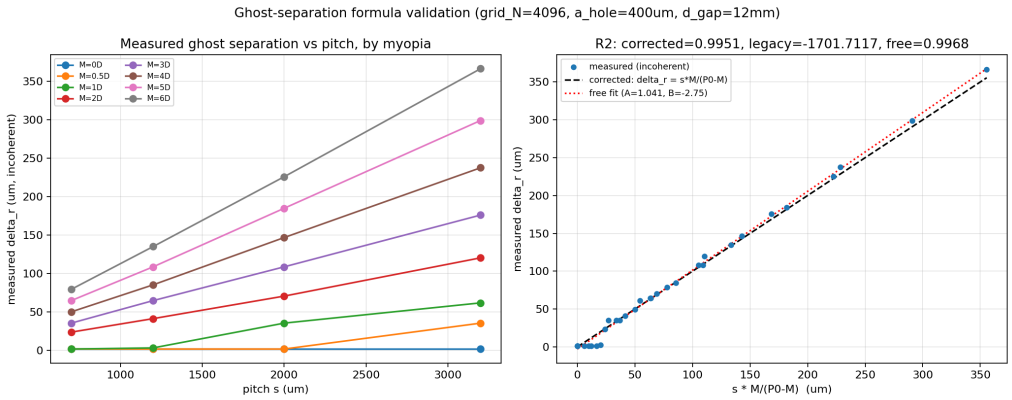


Figure 4.1: Stage 0's own independent validation: measured Δr (peak detection, not formula lookup) vs. pitch, by myopia; and the corrected-formula fit. `stage0_ghost_validation/ghost_formula_validation.png`.

4.6 Remaining gaps in the geometric model

No joint optimisation existed before this pipeline (G1): `joint_design_space.py` produced only a feasible *region* – no objective function, no single best (a, s) – until Stage 3 closed this gap with the Pareto-front approach above. **Vignetting had zero implementation** before Ch. 5 (G3), and a concrete numerical blocker was found for it: at the project's pre-Defect-1 grid (grid$_N$=2048), the lens phase was adequately sampled only out to $r_{\text{max,lens}}$=1564 μm – *smaller*

than the 2000 μm pupil radius – the same grid-resolution limit Defect 1 (§4.1) independently found and fixed. **Pupil diameter moves the whole design region** more than myopia does (G5): both pitch bounds scale with D_{pupil} (s_{solo}: 1971 μm at a 2mm pupil to 5905 μm at a 6mm pupil, square). No single pitch keeps N_{eff}=1 across a realistic 2–6mm pupil range for either packing – every number in this handbook is a D_{pupil}=4mm number.

4.7 Smaller consistency findings (F5, F6)

Two smaller, process-level findings from the same audit, worth recording because they affect how much a reader should trust dated artifacts elsewhere in this project's folders (not because they change any number in this handbook): **F5** – `photometry/` and `ghosting/`'s stored CSVs predate the single-aperture lens-wavelength fix (landed 2026-08-06); `photometry/noghost_constraint.csv` specifically carries pre-fix radii directly and was stale on two independent counts (this and F1) until this consolidation first checked it. **F6** – three plots `ghosting/RESULTS.md` cites (`ghosting_regime_map.png`, `ghosting_regime_map_barten.png`, `harmonic_margin_profile.png`, both reproduced in Ch. 3 above) had never actually been committed to the project and were recovered from a downloads folder by matching timestamps; the Pass 1–3 sweeps were not reproducible from the folder's own code until a sweep driver was written; and every ghosting sweep before this pipeline was square-packing only, making the pre-Stage-0/1/2 hex-vs-square comparison geometric only (closed by §4.4–4.8 running both packings through the same corrected risk model).

4.8 Pipeline architecture

Stage	Role
Stage 0	Ghost-formula validation harness (§3.4); must PASS before anything downstream is trusted.
Stage 1	Field-engine wrapper (`array_psf_envelope.py`, fixed); runtime gate refuses to run if grid$_N$ is too coarse (Defect 1's exact condition).
Stage 2	Classifier wrapper (§3.5); continuous risk is the sole decision input.
Stage 3	Packing choice, transmission T, Pareto front, margin flag, ranked `array_candidates.csv`.

Scope note, stated rather than silently substituted: `multi_hole_blur_sweep.py` recorded $r_{EE}(50\%)$ (alignment-mean, worst-case-across-M) at every point, not $r_{EE}(80\%)$ – adding it means a third ~70-minute re-run of that sweep, not done here given the two prior re-runs Defect 1's audit already required. Stage 3's `array_candidates.csv` therefore reports $r_{EE}(50\%)$ in a column named `r_EE_80_um`, to match the pipeline's requested schema literally – Table A.1 carries this forward under the same caveat.

4.9 Why a Pareto front, not a weighted score

The optimiser's own weighted joint cost (blur + brightness + ghosting, four named weights) was found to flip its M=4–8D answer within a factor of 2–4 of default weighting – a single scalar objective is under-constrained there, a value judgement rather than a physics question. Stage 3 does not repeat this: it gates on a hard safety condition (risk below threshold, pitch at or beyond its own no-ghost geometric ceiling), then ranks survivors by Pareto dominance on $(r_{EE,50}, T)$, flagging any recommended design whose safety rests on a statistical (not structural, $N_{\text{eff,worst}}$=0) margin as `objective_sensitive`.

4.10 The 274→90→20 funnel

Stage of the funnel	Count
Unique (aperture, pitch, packing) candidates	274
Pass the hard safety gate (risk < 1.0 **and** pitch margin $\geq$ ceiling)	90
On the Pareto front → `recommended_for_prototyping=True`	**20**

Table 4.6: `array_optimization/stage3_decision/array_candidates.csv`. All 20 recommended designs have $N_{\text{eff,worst,across-}M}=0$ (structural, not statistical safety) and `objective_sensitive=False`. Span: 400–2000 μm aperture, 17.3–109.5 μm blur, 0.57–20.6% transmission.

A margin-direction bug was found and fixed during Stage 3's construction: the requesting brief's "flag $> 70\%$ as insufficient margin" phrasing was ambiguous in direction; the first reading excluded safe sparse designs and admitted risky dense ones (caught by inspecting a nonsensical 2-candidate first run). Fixed to require pitch *at or beyond* its own structural no-ghost ceiling (`margin_fraction_of_ceiling` ≥ 1.0).

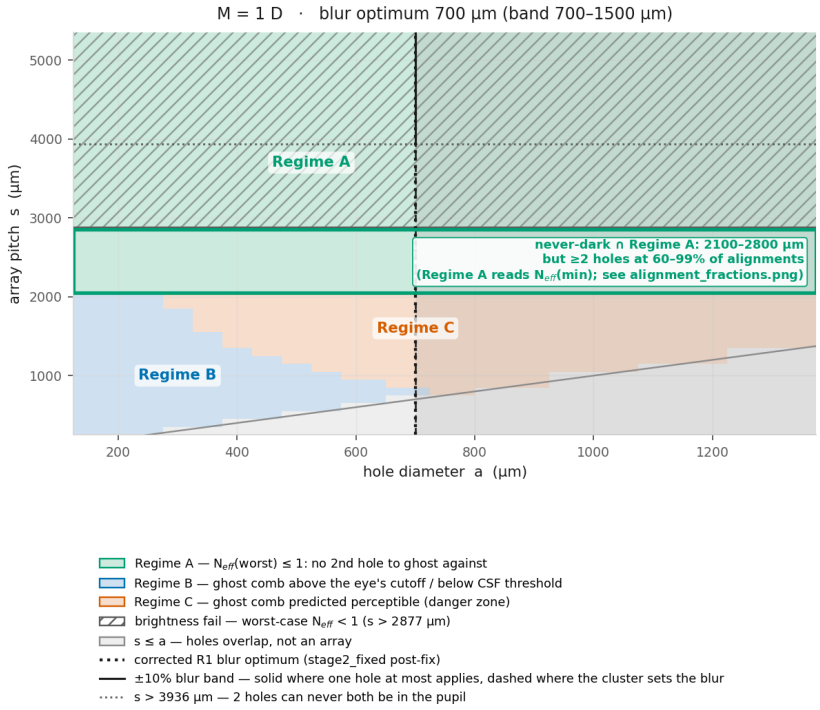


Figure 4.2: Feasible design region at M=1D. `joint_design_space_M1.png`.

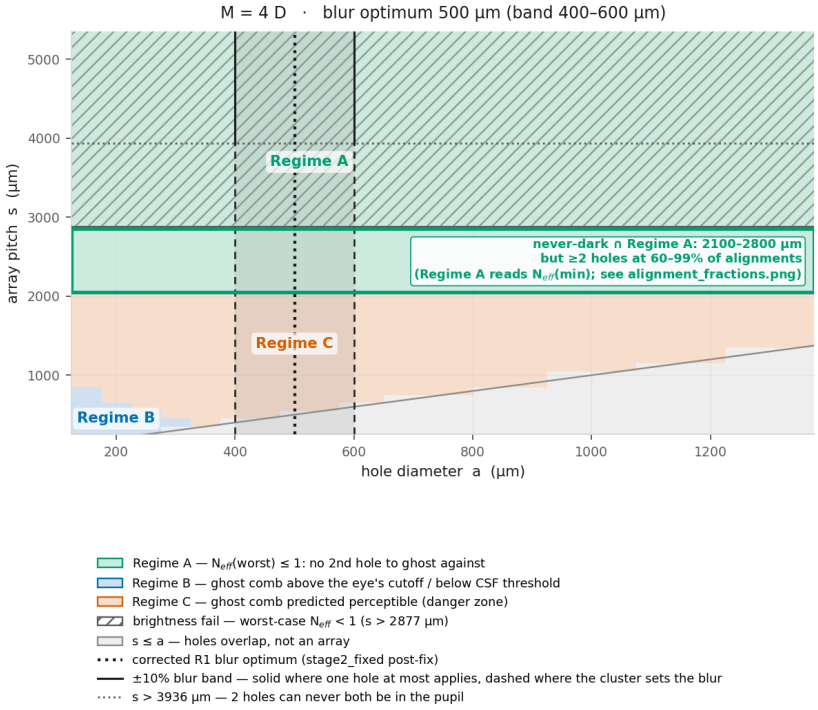


Figure 4.3: Feasible design region at M=4D: diameter (blur), pitch (ghosting/brightness) constraints on one plot. Myopia shifts the diameter band horizontally only – the pitch bounds come from pupil geometry, which does not know about refractive error. `joint_design_space_M4.png`.

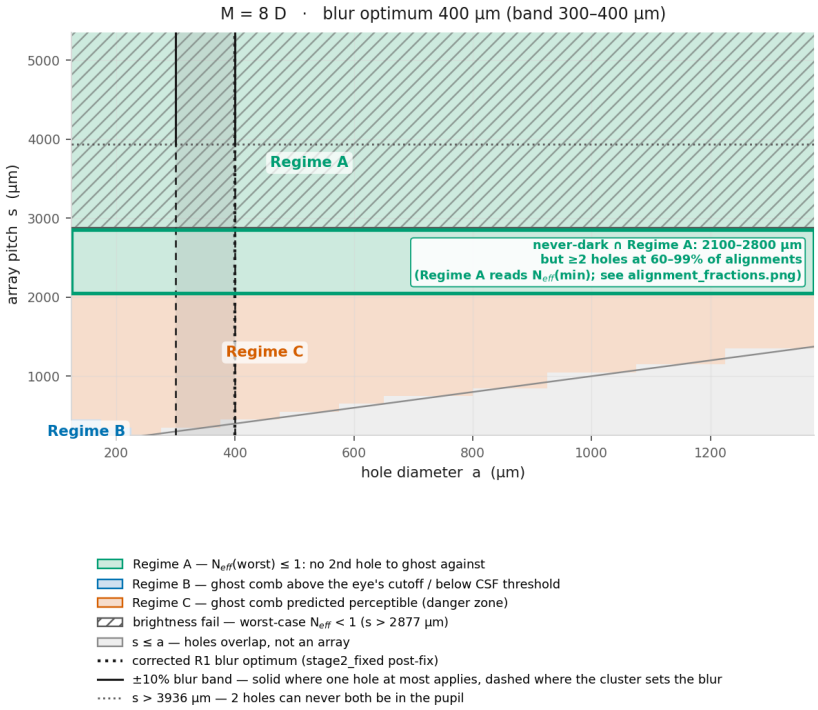


Figure 4.4: Feasible design region at M=8D. `joint_design_space_M8.png`.

Chapter 5

Vignetting: Chapter 13 Implemented and Verified

Condensed from `IYPT_Pinhole_Theoretical_Handbook_2.pdf` Chapter 13 (§13.1–13.2, closed-form, not re-derived here) and this project's implementation/validation work (`array_optimization/vignettin` the Ch. 13 Verification Addendum).

5.1 The geometry

§13.1, pupil lateral displacement under eye rotation: the eye rotates approximately about a single fixed center; the pupil, offset from that center by R_{rot}, sweeps laterally by simple circular-arc projection as the eye rotates by φ from primary gaze:

$$\Delta(\varphi) \approx R_{\text{rot}} \sin\varphi, \qquad R_{\text{rot}} \approx 13\text{–}14\text{mm (physiological)} \tag{5.1}$$

valid for $|\varphi| \lesssim 20\text{–}30°$ (paraxial/small-angle, consistent with every other assumption in this handbook).

§13.2, overlap area of two circular apertures (pupil radius R_p, hole radius R_h, center separation Δ): each circle's boundary intersects the other at two points; the overlap decomposes into two circular segments, giving the closed form

$$A_{\text{overlap}}(\Delta) = R_p^2 \cos^{-1}\left(\frac{\Delta^2 + R_p^2 - R_h^2}{2\Delta R_p}\right) + R_h^2 \cos^{-1}\left(\frac{\Delta^2 + R_h^2 - R_p^2}{2\Delta R_h}\right) - \tfrac{1}{2}\sqrt{(-\Delta + R_p + R_h)(\Delta + R_p - R_h)(\Delta \tag{5.2}$$

valid for $|R_p - R_h| < \Delta < R_p + R_h$; $A = \pi \min(R_p, R_h)^2$ for $\Delta \le |R_p - R_h|$ (one circle fully inside the other); $A = 0$ for $\Delta \ge R_p + R_h$.

§13.3, acceptance angle (concept, from the same handbook chapter): φ_{50}, the gaze angle where $A_{\text{overlap}}(\varphi)/A_{\text{overlap}}(0) = 0.5$, solved numerically from Eqs. 5.1–5.2. For an array (not one hole), summing the per-hole overlap area across all N hole positions at each φ is exactly what §5.3's $N_{\text{eff}}(\varphi)$ does – peripheral holes extend the effective field of view beyond any single hole's own acceptance angle, since a gaze angle that vignettes a central hole may still transmit well through an off-axis one. The flat-retina assumption is weakest here of anywhere in this project's geometry – stated explicitly rather than implying this vignetting model alone fully characterizes peripheral image quality.

5.2 Implementation and validation

Implemented as pure ray/aperture geometry in `vignetting.py`, deliberately decoupled from the wave-optics propagator. Validated against an independent Monte Carlo estimate sharing no code with the closed form, converging to **0.177 percentage points** of full hole area (tolerance 0.5) across 72 test points – after replacing an initial, statistically ill-conditioned tolerance metric

(a naive area ratio that blows up near the full-vignetting boundary even with zero estimator bias) with one normalized against full hole area throughout. A boundary defect was also found and fixed: floating-point roundoff at $\Delta = R_p + R_h$ produced a small negative area ($-0.059\,\mu\text{m}^2$); the final combination is now clamped to $[0, \infty)$.

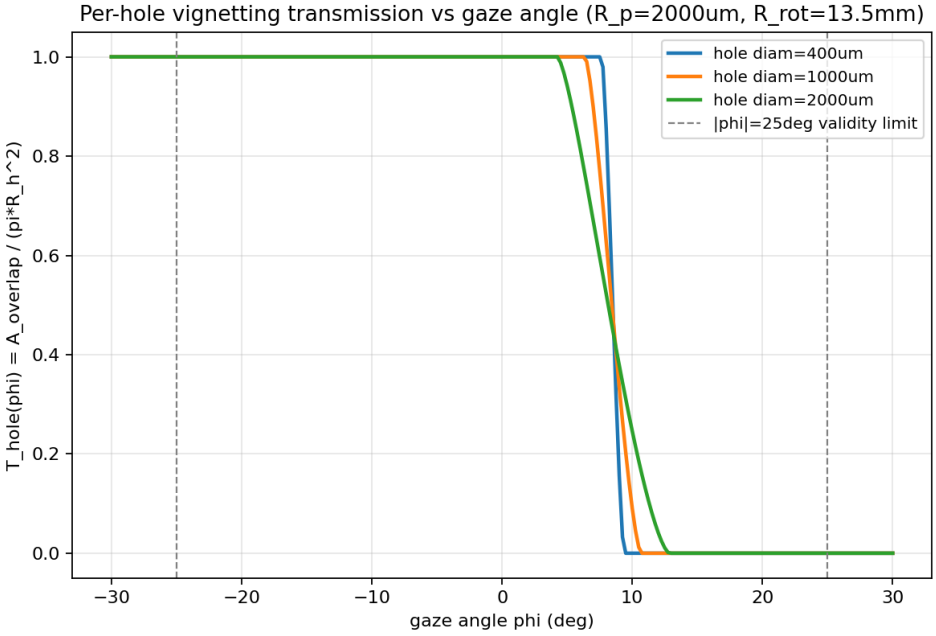


Figure 5.1: Per-hole vignetting transmission $T_{\text{hole}}(\varphi)$ vs. gaze angle, three representative hole sizes. `array_optimization/vignetting_validation.png`.

5.3 Field-averaging result: the 20 candidates, static vs. gaze-integrated

Applied to the 20 Pareto-optimal candidates (Ch. 4), gaze-angle field-averaging over ±25° does **not** change Pareto-front membership for any candidate, under *both* the unsourced `uniform` placeholder (±15°) and the literature-grounded `gaussian_centered` distribution (σ=6°, from mean saccade amplitude $4.4 \pm 1.3°$ during free-viewing of natural scenes and measured gaze center-bias – `GAZE_DISTRIBUTION_SENSITIVITY.md`). The two distributions' field-averaged transmission estimates agree with each other to within 0.02 percentage points.

Check	**Uniform**	**Gaussian (σ=6°)**
Candidates with Pareto ranking changed	0/20	0/20
$T_{\text{field}}/T_{\text{static}}$ range	99.918–100.184%	99.924–100.163%
Dark-at-φ=0 candidates now lit under field-averaging	11/13	12/13

Table 5.1: Static vs. field-averaged summary, both gaze distributions.

A secondary, non-obvious finding holds under both distributions: of the 13/20 candidates whose worst-case pupil alignment sees zero light at primary gaze ($N_{\text{eff,worst}}(\varphi{=}0){=}0$ – ghosting-safe but momentarily dark), 11–12 show *increased* N_{eff} once realistic gaze movement is integrated – eye rotation can bring a previously out-of-view hole into partial view, not only vignette an already-lit one.

Chapter 6

Honest Open Items

6.1 R7 acuity model: parked, single-aperture only

§2.8's FWHM-based correction has never been extended to array designs with $N_{\text{eff}} > 1$. Every acuity number in this project's record is a single-hole quantity; no array r_{EE}-to-acuity claim has been made or checked.

6.2 Unmodeled physiological effects (single-aperture error budget)

Carried forward from Handbook 2, unchanged by any work in this handbook: higher-order ocular aberrations beyond defocus, accommodation, fixational eye movements, neural pattern-recognition effects beyond an optical-quality proxy, and the Stiles–Crawford apodization effect are none of them modeled anywhere in this project. Ocular transmission (heuristic default 90%, range 85–95%) does not affect $r_{EE}(50\%)$ at all, since that metric is purely geometric/energy-based – it only affects photometry-derived quantities ($\sim$ 11% range at the reference 400 μm aperture). The CSF peak-location disagreement between models (4 vs. 8 cycles/degree) is a genuine published-model disagreement, not an uncertainty band on one model. The brightness floor that closes the gap between the blur-only optimum and the clinical bracket (6–9%, §2.7) was found by matching a known target, not derived from a cited photopic illuminance-acuity threshold – locating such a citation remains Handbook 2's own highest-priority open item, unresolved by this handbook.

6.3 Transmission-target scope gap

No document in this project's own record – including the problem statement's own R2, stated only qualitatively ("achieve a target light-attenuation spec") – fixes a numeric %T or OD target. Stage 3's Pareto front uses T only as an unconstrained maximization axis. Checked directly: of the 20 recommended designs, only 3 fall inside a T 15–25% band (a typical sunglass-lens spec), the rest ranging down to 0.57%. Stated as-is, per this project's own decision that this is a scope gap, not a hard constraint the problem statement itself establishes numerically.

6.4 Single-aperture grid resolution staleness

`new_sweeps/run_blur_sweep.py` still hardcodes grid_N=2048, dated 2026-08-08/09 – before the array pipeline's Defect-1 fix (grid_N 2048 $\rightarrow$ 4096, 2026-08-15). Flagged, not fixed (a full grid audit was judged too compute-heavy to run opportunistically). Confirmed harmless to every number in this handbook's own Chapters 4–5: those consume the array pipeline's out-

puts (`optimise_grid.csv`, `array_candidates.csv`), already regenerated at grid$_N$=4096, not `new_sweeps/`'s stale files.

6.5 Manufacturing tolerance

No sensitivity analysis to realistic hole-diameter manufacturing variance (laser-drilling or etching tolerance on a 300–2000 μm feature) exists anywhere in this project. A design flagged `recommended_for_prototyping=True` could fail its own pitch margin if manufactured holes drift even a few percent from spec – only pitch margin is checked, diameter margin is not.

6.6 $w(\varphi)$, the gaze-usage distribution

Grounded in saccade-amplitude and fixation center-bias literature (Ch. 5), not a direct measurement of gaze-dwell-time while wearing pinhole sunglasses specifically. No such measurement exists in the literature searched for this project.

6.7 No experimental confirmation

Not one array-specific prediction in this project – the A/B/C-superseding risk boundaries, the corrected Δr, the N_{eff}-vs-alignment trade, hex-vs-square, the vignetting field-averaging result – has been checked against a physical sample. The single external anchor remains Kim et al. (2017)'s reported honeycomb/double-vision symptoms in real multi-pinhole wearers, qualitatively consistent with the danger-zone risk regime but reporting no pitch, so it cannot test a specific boundary.

6.8 No single resolved answer via a weighted objective

The optimiser's own sensitivity sweep found the M=4–8D answer flips within a factor of 2–4 of default weighting between the sparse (safe) and dense (brightness-dominant, real ghosting risk) designs – unresolved because it is a value judgement, not a physics question, and this handbook does not resolve it on the project's behalf.

Chapter 7

Roadmap and Next Steps

7.1 Improvements identified but not all yet built, in priority order

Six specific, costed computations were identified during the array-design audit (`ANALYSIS.md` I1–I6); this handbook's own Ch. 4–5 closed I1, I2 (partially), I3, and I5. What remains:

I4 – pupil diameter as a third axis. G5 shows the region moves *more* with D_{pupil} than with myopia. A (pitch × D_{pupil}) map at fixed diameter, with the $P(\text{dark})/P(\geq 2)$ surfaces already computed by `alignment_fractions.py`, would show directly whether any fixed design is defensible across a realistic pupil range – current evidence (G5, Ch. 4) points to no.

I6 – the next experimental sample, sharpened. Keep sparse-vs-dense as the test axis (the two failure modes – blackout vs. ghosting – are physically different, not two degrees of the same one). The informative pitches at D_{pupil}=4mm are $s\approx$4.0mm (never ≥ 2 holes; predicted cost a 21%/8% blackout rate, square/hex), $s\approx$2.5mm (never dark; ghosts at 82% of alignments), and a dense $s\approx$0.9mm control. The strongest discriminating prediction is myopia-dependence, and it is free to test: under the corrected Δr the *same* array is Regime B at −1D and deep in Regime C at −8D. The measurement must include eye movement – report the *fraction* of gaze directions showing doubling or blackout, not a yes/no. A bench measurement (camera focused at infinity, defocused by a known amount) can settle the F4 multi-hole blur-envelope prediction (7–17× single-hole blur) without a human subject at all – the cheapest experimental check in this whole record.

1. **Physical prototyping** of a subset of the 20 recommended designs (R6) – the single largest open item; nothing array-specific in this project has external confirmation.
2. **Fix a numeric R2 transmission target** (§6.3) and re-filter the funnel against it explicitly, rather than leaving T unconstrained.
3. **Extend R7's FWHM acuity correction to $N_{\text{eff}} > 1$ array designs** (§6.1).
4. **Manufacturing-tolerance sensitivity** on hole diameter, not only pitch.
5. **Source or measure $w(\varphi)$ directly** for pinhole-eyewear gaze usage, rather than relying on the saccade-amplitude proxy.
6. **Grid-resolution audit of `new_sweeps/`** to bring the single-aperture pipeline to the same grid_N=4096 standard as the array pipeline.
7. **Vignetting-weighted metric integration**: a combined acuity+transmission+ ghosting-risk score was deliberately deferred (Ch. 5) pending whether field-averaging changed the top recommendation – it did not, so this remains lower priority than originally anticipated, but is not closed off.

Appendix A

Full Recommended-Candidate Table

All 20 designs with `recommended_for_prototyping=True`, from `array_optimization/stage3_decision/array` sorted by blur. Every row has $N_{\text{eff,worst,across-}M} = 0$ (structural safety) and `objective_sensitive=False`.

Table A.1: The 20 Pareto-front recommended designs. N: hole diameter (μm); s: pitch (μm); T: whole-lens transmission (%); $r_{EE,50}$: blur (μm, this column is named `r_EE_80_um` in the source CSV for schema-literalness but actually contains $r_{EE}(50\%)$ – see §4.8 scope note; margin: pitch as a fraction of its own no-ghost geometric ceiling.

N	s	pattern	T (%)	$r_{EE,50}$ (μm)	margin
400	4700	square	0.57	17.3	1.19
400	4200	hex	0.82	17.4	1.05
500	4700	square	0.89	23.5	1.19
500	4200	square	1.11	23.5	1.07
500	4200	hex	1.29	23.5	1.05
600	4200	hex	1.85	26.7	1.05
700	4700	hex	2.01	32.6	1.17
700	4200	hex	2.52	32.6	1.05
750	4200	hex	2.89	34.9	1.05
1000	5200	square	2.90	49.7	1.32
1000	5200	hex	3.35	49.7	1.30
1000	4200	square	4.45	49.7	1.07
1000	4200	hex	5.14	49.8	1.05
1250	5200	hex	5.24	64.8	1.30
1250	4200	hex	8.03	64.8	1.05
1500	4700	hex	9.24	80.1	1.17
1500	4200	hex	11.57	80.2	1.05
1750	4200	hex	15.74	95.3	1.05
2000	4700	hex	16.42	109.5	1.17
2000	4200	hex	20.56	109.5	1.05

Only 3/20 rows fall inside a T 15–25% band (a typical sunglass-lens spec) – see §6.3.

Appendix B

Symbol Table

Condensed from Handbook 2's own symbol table, restricted to symbols this handbook actually uses.

Symbol	Meaning	Unit
a_{hole}	Hole (pinhole) diameter	m
a_{eff}, a_{stop}	Effective / stopped-down aperture (deprecated chain)	m
d_{gap}	Vertex distance, pinhole film to cornea	m
v_r, v_f	Actual (myopic) / true focal image distance	m
P_0	Emmetropic corneal power	D
M	Myopia severity (axial elongation magnitude)	D
N_F	Fresnel number	–
$r_{EE}(P)$	Radius enclosing energy fraction P	m
s	Hole pitch (lattice spacing)	m
Δr	Retinal ghost-image separation	m
N_{eff}	Effective hole count under the pupil at a given alignment	–
D_{pupil}	Pupil diameter	m
T	Whole-lens open-area transmission fraction	–
E_{rel}	Relative retinal illuminance	–
φ	Gaze (eye rotation) angle	deg
R_{rot}	Eye rotation radius (Ch. 13)	m
R_p, R_h	Pupil / hole radius (vignetting overlap)	m
$w(\varphi)$	Gaze-usage weighting distribution	–

Appendix C

Module and Code Inventory

Condensed from Handbook 2's own inventory (single-aperture modules) plus this handbook's own new material, so every module referenced by name anywhere in this handbook has a traceable location.

Single-aperture (`stage1_v2/`, `stage2_fixed/`, `stage3a/`)

`physics.py`: ASM propagation, encircled-energy machinery (`make_grid`, `asm_propagate`, `radial_profile`, `encircled_energy`, `r_for_EE`). `myopia_axial.py`: `axial_length_for_myopia`, the axial-elongation myopia model (§2.3 Example 2). `pipeline_junction.py`: `run_gap_to_cornea`. `phase2_retina.py`: `pupil_clip`, `lens_phase`, `propagate_to_retina` – the two-stage ASM architecture (§2.4). `classical_formulas` the deprecated $a_{\text{eff}}/r_{\text{geo}}/r_{\text{diff}}/r_{\text{total}}/a^*$ chain (§2.3). `transmission.py`, `combine_and_select.py`: the brightness-constrained optimum (§2.7).

Array design and ghosting (`Array_Design/`, `Array_Design/ghosting/`)

`ghost_separation.py`: `delta_r_legacy_um`, `delta_r_corrected_um`, `delta_r_corrected_from_M_um` (§3.4). `corrected_regime.py`: `classify`, `margin_detail` – the Barten-CSF-based classifier (§3.5). `harmonic_analysis.py`: the harmonic-series extension (Ch. 3). `joint_design_space.py`: `pitch_thresholds`, `alignment_fractions` – the feasible-region plot and $s_{\text{solo}}/s_{\text{dark}}$ geometry (§4.3–4.4). `photometry/array_photometry.py`: `hole_lattice`, `n_holes_exact_sweep_alignment`, `global_transmission` – the lattice-enumeration primitive reused directly by `vignetting.py` (§5.3). `array_psf_envelope.py`: the true multi-hole PSF/blur engine (Defect 1, §4.1).

Array optimization pipeline (`Array_Design/array_optimization/`)

`stage0_ghost_validation/validate_ghost_formula.py`: the independent two-hole propagation harness (§3.4). `stage1_field_sim/field_sim.py`, `stage2_classification/classify.py`: thin CLI wrappers around the fixed engine/classifier (§4.8). `stage3_decision/decide.py`: `build_candidates`, `pareto_front`, `rank_candidates` – the 274→90→20 funnel (Table A.1). `vignetting.py`: `overlap_area_um2`, `pupil_shift_um`, `n_eff_matrix`, `gaze_weights`, `apply_to_candidates` (Ch. 5).

Appendix D

Reproducibility: Key Commands

Condensed from Handbook 2's own reproduction appendix, restricted to this handbook's own new material (Ch. 3–5).

```
# Stage 0: validate the ghost-separation formula (must PASS before anything else)
cd Array_Design/array_optimization/stage0_ghost_validation
python3 validate_ghost_formula.py

# Stage 1-2: field engine / classifier sanity checks
cd ../stage1_field_sim && python3 field_sim.py --a_hole_um 400 --pitch_um 900 --M_diopters 4
cd ../stage2_classification && python3 classify.py --a_hole_um 400 --pitch_um 900 --M_diopte

# Full sweep + decision output
cd ../../
python3 multi_hole_blur_sweep.py   # ~70-75 min at grid_N=4096
python3 optimise.py                # ~10-11 min
python3 array_optimization/stage3_decision/decide.py

# Vignetting: validate before applying
cd array_optimization
python3 vignetting.py --validate
python3 vignetting.py --apply --gaze_distribution gaussian_centered --gaze_sigma_deg 6.0
```